

CEO Individualism and the Distribution of Performance Risk

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Abstract

I measure CEO individualism using only the CEO's answers to analyst questions on earnings calls—answers delivered without a prepared script, and so less filtered than the remarks that open the same call—projecting each sentence onto a transformer-based individualism axis built from 100 validated anchor sentences and dropping answers to narrow numerical and procedural questions. In an ExecuComp firm-year panel, 2010–2023, a one-standard-deviation increase in CEO individualism raises the pass-through of a one-unit Δ ROA shock to employment by 7.3 percentage points. The effect is concentrated in downturns: individualistic CEOs cut workforces aggressively, while the good-times differential is positive but imprecisely estimated. I call this a *scarcity-revealed weighting* channel: the relative weight a CEO places on shareholder returns versus employee welfare is observable from outcomes only when scarcity forces a binding tradeoff. The asymmetry favors this channel over a symmetric efficiency story, though equality of the two interactions is rejected only at the 10% level. A separation-composition margin agrees: in downturns, more of the separations at individualistic CEOs' firms are of the abrupt kind least likely to be voluntary, though the same tilt appears more weakly in upturns, marginally significant rather than a clean null.

Keywords: CEO individualism; managerial style; labor margin; performance risk; transformer-based text measure; earnings calls.

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1 Introduction

When a firm’s performance falls, who bears the cost—shareholders or employees? The allocation is not contractually pinned down; it is the CEO’s to make. This paper shows that one dimension of CEO disposition—individualism, measured from earnings-call language—systematically shifts more of that cost onto employees; only the employee margin is measured here, and no shareholder-side outcome is estimated. A one-standard-deviation increase in CEO individualism raises the pass-through of a one-unit Δ ROA shock to firm employment by 7.3 percentage points, roughly doubling the 7.6-point pass-through of the average CEO in the sample (Table 4, Column (4): $\beta_2 = 0.0730$, $\beta_1 = 0.0755$); outside 2020–2021 that average falls to 4.3 points and is insignificant while the individualism differential strengthens (Table 10, Column (1)). That differential is concentrated in downturns: individualistic CEOs cut workforces aggressively when performance falls, while the upside interaction is positive but imprecise (+0.033, $t = 1.16$) and equality of the two interactions is rejected only at the 10% level ($p = 0.086$; Table 5).

I refer to this as a *scarcity-revealed weighting* channel: the relative weight a CEO places on shareholder returns versus employee welfare is observable from outcomes only when scarcity forces a binding tradeoff. Under positive performance shocks the firm’s surplus expands and the allocation between constituencies is largely non-rivalrous, so CEO objective weighting has limited differential consequences for outcomes; under negative shocks the loss must be borne by some constituency, and CEO weighting determines whether shareholders or employees absorb it (Cornell and Shapiro, 1987; Hart and Zingales, 2017)—an asymmetry a symmetric efficiency mechanism does not predict.

The labor margin is the sharpest setting in which to test the channel: labor expense is quasi-fixed and does not contract proportionally with revenue (Anderson et al., 2003; Banker and Byzalov, 2014), and the wedge between sticky labor cost and cyclical revenue is a central determinant of operating leverage (Donangelo, 2014; Donangelo et al., 2019). I measure the quantity side of that expense, headcount.

Two innovations underpin the estimate. *First*, I introduce a transformer-based measure of CEO individualism (“Indiv”) built from the 364,222 ExecuComp earnings-call transcript records (2008–2024) that carry at least one scorable answer to a substantive analyst question (Table 2); grouped by CEO-quarter, these records become 122,721 cells (Table 1, Step 2). For each sentence a CEO speaks, I compute its position on an individualism axis built from 100 validated anchor sentences, screened for emotional, performance, and capability contamination and residualized against a first-person pronoun direction—using a pretrained sentence-Transformer encoder (Reimers and Gurevych, 2019). I score only the CEO’s answers to substantive analyst questions, setting aside answers to purely numerical modeling questions and to clarifying or procedural ones, which speak to the analyst’s spreadsheet rather than to the manager. The measure is stable (split-half $\rho \approx 0.81$ on the data-sufficient subsample). Its low correlation with the Chatterjee and Hambrick (2007) pronoun ratio ($\rho \approx 0.18$ at the CEO-quarter level) shows that the residualization holds against an externally constructed benchmark. The disposition it captures travels with the person more than with the firm: for CEOs observed at two different firms the two firm-level scores correlate at +0.672, and CEO fixed effects explain a further 10.6% of the variance in the measure beyond firm fixed effects, against 0.76% in the other direction (Table 12, Column (1)). The measure is taken from answers delivered without a prepared script, and so less filtered than the remarks that open the same call: each firm-year’s Q&A score differs systematically from the score computed on the prepared section of the same calls (paired $t = 229.7$ over 29,734 firm-years), and when both scores enter the employment regression together only the Q&A score’s interaction with ΔROA is statistically significant (+0.062, $t = 2.26$); the prepared-remarks interaction is not (+0.036, $t = 1.22$).

Second, I supplement standard fixed effects with a saturated specification that absorbs a firm-specific slope on ΔROA (Firm $\times$ c. ΔROA), so that the coefficient of interest is identified only from within-firm changes in CEO individualism that move the firm’s ΔROA –employment slope. This addresses the leading endogeneity concern in CEO-style work—time-invariant matching between individualistic CEOs and intrinsically more

cyclical firms. Under saturation, restricted to firms with at least five firm-years, the point estimate is 0.081 ($t = 2.21$; Table 10, Column (2)) on a smaller sample than the 0.073 baseline; its survival, not its size, is what endogenous matching would not predict. For that baseline estimate, unobserved selection would have to be 23.8 times as important as the full control set to drive the interaction to zero (notes to Table 4). The baseline estimation sample is 13,376 firm-years of nonfinancial, nonutility firms over 2010–2023.

The asymmetric decomposition above is the pattern scarcity-revealed weighting predicts: $\hat{\beta}^- = +0.116$ ($t = 3.55$). Four further checks—run on overlapping samples, not independent ones—point the same way. First, controlling directly for the two observable firm-culture channels—the firm’s scripted communication and the individualism of strictly prior CEOs at the same firm, the terms isolated in Table 11—moves the interaction from +0.082 to +0.059 on the prior-CEO subsample and leaves it significant ($t = 1.99$; Table 9, Columns (3) and (5)), though no test separates it from the channel terms. Second, as noted above, excluding the COVID-19 period strengthens the interaction to +0.093, and in a horse race against the Chatterjee and Hambrick (2007) pronoun ratio the Indiv interaction is +0.075 ($t = 4.02$) while the pronoun-ratio interaction is negative and insignificant (-0.011 , $t = -0.56$). Third, the pattern holds in the panels of firms whose CEO changed for reasons plausibly unrelated to firm prospects—firm samples, not event windows: in firms with a death- or health-driven succession (66 firms, 604 firm-years; Table 6, Column (4)) the downside interaction stays positive and significant (wild-cluster bootstrap $p = 0.011$, concentrated in, though not driven solely by, energy successions), and in firms with a retirement-driven succession by a CEO aged 60 or older (783 firms, 7,948 firm-years) its magnitude (+0.116, $t = 2.62$; wild-cluster $p = 0.007$; Table 7, Column (2)) matches the full-sample downside estimate. Fourth, in individual-level position records the share of separations not followed by a new position within ninety days rises more steeply in downturns for firms run by individualistic CEOs (-0.0612 , $t = -4.61$, on the weakly negative ΔROA^- ; Table 8, Column (1)), and a Wald test rejects linearity in ΔROA ($p < 0.001$); the upside gradient is marginally significant (+0.0260, $t = 1.95$) rather than a clean null, and no test compares the two magnitudes.

The paper sits within an emerging CEO-level cultural-finance literature. Country-level work built on the Hofstede (2001) individualism index documents links to momentum profits and overconfidence-driven trading (Chui et al., 2010), to cross-border M&A volume and synergy gains (Ahern et al., 2015), and to risk perception (Wu and Zhong, 2025); An et al. (2025) push the cultural lens inside the firm by proxying CEO individualism with first-name uncommonness and linking it to financial misstatements.

The paper makes three contributions. First, a transformer-based measure of CEO disposition that complements rather than supplants pronoun-based proxies and that extends finance text-measurement (Loughran and McDonald, 2011; Hassan et al., 2019) from sentiment to disposition. Second, an extension of the CEO-style literature (Bertrand and Schoar, 2003; Custódio et al., 2013; Graham et al., 2013; Schoar and Zuo, 2017) from investment, financing, and disclosure into the labor margin and the distribution of performance risk. Third, the *scarcity-revealed weighting* channel itself: a tradeoff-binding-under-scarcity logic that discriminates a CEO-disposition mechanism from a symmetric efficiency story. The discriminating evidence is the asymmetric decomposition of Table 5; the saturated firm-slope specification estimates the symmetric interaction only.

2 Hypotheses Development

CEO disposition is not observed directly. I infer it from the language CEOs use in the question-and-answer portion of quarterly earnings calls—answers delivered without a prepared script, and so less filtered than the remarks that open the same call—using the Indiv score constructed in Section 3. The employment margin is the lens: workforce adjustment is the most direct and economically consequential channel through which CEOs distribute performance shocks between shareholders and employees.

2.1 Theoretical Anchors

Psychological substrate: self-construal. The individualism–collectivism dimension has deep roots in cross-cultural psychology (Hofstede, 2001; Triandis, 2001) and provides the conceptual basis for the finance literature that links individualism to overconfidence, trading, and corporate behavior (Chui et al., 2010; Ahern et al., 2015; An et al., 2025). At the individual level, Markus and Kitayama (1991) distinguish between an *independent* self-construal—in which the self is bounded, agentic, and defined by personal goals—and an *interdependent* self-construal, in which the self is contextual and defined through in-group relationships. Independent self-construal places lower weight on the welfare of relational others, including employees, when the focal decision involves resource allocation across constituencies. The individualism–collectivism continuum I measure is intended as a description of how a CEO weighs employees as stakeholders, not of how fast she acts. The two are separated only in part: the asymmetry test below is designed to separate stakeholder weighting from decisiveness that operates symmetrically—and rejects equality only at the 10% level—but not from a speed advantage specific to negative signals, which is retained among the alternative readings.

Economic mechanism. The shareholder–employee tradeoff is most consequential—and most directly observable—on the labor margin (Cornell and Shapiro, 1987; Hart and Zingales, 2017). A central feature of this tradeoff is that it does not bind uniformly across performance states. Under positive performance shocks, the firm’s surplus expands and the allocation between constituencies is largely non-rivalrous: shareholders and employees can both benefit. Under negative performance shocks, by contrast, the loss must be borne somewhere, and the CEO’s relative weighting of shareholder returns versus employee welfare shapes how much of it is passed through to employees through workforce reductions rather than absorbed on the shareholder side; only the employee margin is measured here. I refer to this mechanism as the *scarcity-revealed weighting* channel: the CEO’s underlying weighting—the heterogeneous weight different CEOs place on competing constituencies—is *revealed* by adverse shocks because that is when

the tradeoff binds, and the individualism score is intended as a proxy for it, subject to the alternative readings set out below.

The link between individualism and tradeoff resolution operates through two complementary psychological channels. (i) *Internalization of relational obligations*. The lower internalization of social norms that makes individualistic CEOs more willing to pursue self-defined objectives over group consensus (Hofstede, 2001; Triandis, 2001) also makes them less constrained by relational considerations toward the firm's workforce when resolving allocation tradeoffs. (ii) *In-group framing of employees*. The Markus and Kitayama (1991) interdependent self-construal frames employees as members of an in-group whose welfare is part of the self-relevant objective; the independent self-construal frames them as resources external to the self. Both channels therefore predict that an independent self-construal weights shareholder returns more heavily relative to employee welfare when allocation tradeoffs bind.

Normative neutrality. Individualism and collectivism are value orientations, not ranked traits. Collectivist commitment to workforce stability protects employees but can sustain inefficient labor allocations; individualist willingness to adjust can enforce resource discipline but can erode firm-specific human capital. The paper documents a behavioral regularity, not a welfare ranking.

2.2 Hypotheses

H1 (pass-through). *The pass-through of operating-performance shocks to firm employment is increasing in CEO individualism.*

H1 alone, however, is consistent with two distinct mechanisms that have very different welfare and pricing implications.

Mechanism A (efficiency). Individualistic CEOs are simply more decisive resource allocators: they move labor up and down faster in response to performance signals, reducing adjustment-cost-driven sluggishness in both directions. Under this account,

the pass-through of performance shocks to employment is higher under individualistic CEOs in *both* expansions and contractions.

Mechanism B (scarcity-revealed weighting). Individualistic CEOs do not symmetrically improve labor responsiveness; rather, they place greater weight on shareholder returns relative to employee welfare *when scarcity forces a binding tradeoff*. The asymmetry follows from the structure of the tradeoff set out above: objective weighting has differential consequences for outcomes only when the loss must be borne by some constituency. Under this account, the pass-through differential is concentrated in downturns and is absent—or much smaller—in upturns.

The two mechanisms are observationally equivalent in a symmetric specification but generate sharply different predictions once positive and negative performance shocks are separated. This delivers the asymmetry hypothesis, the central mechanism prediction. Statistically distinguishing the two requires sufficient power on *both* sides of the shock distribution, and a precise rejection of equality is not guaranteed even when the point-estimate pattern is informative; I return to this caveat in the empirical analysis.

H2 (asymmetry). *The CEO-individualism differential in employment pass-through is concentrated in negative performance shocks; the differential is absent or substantially smaller in positive shocks.*

H3 (composition). *In downturns the individualism differential also appears in the composition of separations, not only in the size of the employment adjustment: the involuntary share of separations rises more steeply under individualistic CEOs; the differential is substantially smaller in upturns.*

Empirical implication. In a regression of $\Delta \log(\text{Emp}_{i,t})$ on $\Delta \text{ROA}_{i,t}$ interacted with the CEO-individualism score $\text{Indiv}_{i,z}$, H1 predicts $\beta_2 > 0$; decomposing $\Delta \text{ROA}_{i,t}$ into non-positive and non-negative components, H2 predicts that the interaction coefficient on the non-positive component exceeds that on the non-negative component. The same decomposition run on the involuntary separation share delivers H3, but with the predicted sign reversed: since $\Delta \text{ROA}^- \leq 0$ and a harsher adjustment raises rather than lowers that

share, H3 predicts a *negative* coefficient on the non-positive component. Support for H1 and H2 jointly favors the scarcity-revealed weighting channel over the efficiency mechanism. H3 does not separate the two mechanisms; it locates the incidence. H1 is tested in Table 4, H2 in Table 5, and H3 in Table 8.

Alternative readings. The asymmetric pattern is also consistent with at least three alternative interpretations that the present design cannot fully separate from the scarcity-revealed weighting channel: (a) heterogeneity in CEO *risk tolerance* on the labor margin, (b) heterogeneity in *compensation incentives* that tie CEO pay to short-run profitability and thereby encourage downturn layoffs, and (c) heterogeneity in *information processing* that allows individualistic CEOs to act faster on negative signals. I treat the objective-weighting reading as the most natural framing given the anchors above, and revisit the alternatives in the limitations.

3 Data and Method

3.1 Sample

I construct a firm-year measure of individualism, *Indiv*, from earnings call transcripts. The text data come from the WRDS S&P Capital IQ Transcripts database for fiscal years 2008 through 2024. I restrict the sample to firms covered by ExecuComp—which provides verified CEO names, annual CEO flags, and firm linkages via *gvkey*—which the CEO-speaker match of Section 3.2 requires. After this restriction the raw sample contains 376,002 transcript records, of which 364,222 carry at least one CEO answer to a substantive (C1–C3) analyst question; those records are the scoring universe used throughout (Table 2). The unit counted throughout is the transcript *record*, not the call event: the database publishes several versions of each transcript (spellchecked, proofed, edited, audited) and I do not de-duplicate them, so the 364,222 records cover 114,627 distinct call events, a median of three records each, and the median firm-year carries thirteen scored records over four call events. Grouped by speaker and calendar quarter, the same records form the 122,721 CEO-quarter cells used for the reliability

tests of Table 2 and reported as Step 2 of Table 1; the firm-year measure itself is built from record-level scores directly (Section 3.6). At the firm-year stage I drop financial firms and utilities using the Fama–French 48 industry classification, following standard practice in the cost-stickiness literature (e.g., [Anderson et al. \(2003\)](#))—the largest firm-level cut, 8,460 of the 30,058 firm-years that reach Compustat (Table 1, Steps 6–7)—and require CRSP–Compustat coverage: a valid PERMNO–gvkey linkage and non-missing Compustat fundamentals. That coverage requirement is a data-coverage filter rather than a CEO-related one, reducing the panel from 21,598 to 19,076 firm-years (Table 1, Steps 7–8). The panel narrows to 13,376 firm-years in the main specification with the full set of firm- and CEO-level controls (Table 4, Column (4)), covers fiscal years 2010–2023 (after the lag-construction window), and spans 38 of the 43 Fama–French industries that survive the financial and utility exclusion (untabulated). Those 13,376 are 44.5% of the 30,058 scored firm-years that reach Compustat, and the direction of that attrition is untested.

Table 3 describes that estimation sample. Two features of it bear on what follows. First, the performance shock is close to symmetric in the cross-section—its non-positive and non-negative segments have means of -0.026 and $+0.027$ —so the asymmetry documented in Table 5 is a difference in *response*, not an artifact of how often firms are in each regime. Second, $\text{Indiv}_{i,z}$ is not orthogonal to the controls: it correlates -0.40 with lagged labor intensity and $+0.18$ with firm size, so more individualistic CEOs are observed at larger and less labor-intensive firms. Firm and industry fixed effects absorb the time-invariant part of that association, and the coefficient of interest is an interaction rather than a level, but the correlation is why the saturated and succession designs of Section 4 are reported rather than the baseline alone. Lagged labor intensity and the lagged labor share are themselves correlated at 0.51 , which is expected given that the latter is built from an imputed labor expense. The [Gentry et al. \(2021\)](#) turnover database, Revelio Labs *Positions* records and the Equilar CEO panel are used later and described in the notes to the tables that use them. Two of the controls defined in Table 13 are proxies: CEO tenure is a left-truncated ExecuComp appearance count, and the lagged

employment labor share (ELS) is built on an imputed labor expense.

3.2 Identifying the CEO Speaker

For each transcript I identify the CEO’s `transcriptpersonid` in the call using a score-based heuristic that combines five signals: (i) the number of times the speaker is addressed by the speaker’s first or last name in analyst questions, (ii) standard handoff phrases (e.g., “*turn it over to John*,” “*hand the call to Smith*”), (iii) explicit self-introductions (“*I am John Smith, CEO*”), (iv) a penalty for self-thanks patterns that typically identify CFOs or IR personnel rather than the CEO, and (v) speech length measured in words. Each candidate `transcriptpersonid` receives a composite score; the highest-scoring candidate is assigned as the CEO and given a confidence rating (High / Medium / Low / Unreliable) based on the magnitude of supporting evidence. The rating is recorded but is not used as a filter: every record with an ExecuComp-matched CEO speaker enters the scoring universe whatever its rating, and High- and Medium-confidence records are 42.0% of that universe (untabulated). The measure is therefore built on the full range of identification confidence, and no specification in the paper restricts it.

3.3 Sentence-Level Corpus and Filtering

For each identified CEO and each transcript, I extract the full set of sentences spoken by the CEO, separating them into prepared remarks (`transcriptcomponenttypeid` = 2) and Q&A responses (`transcriptcomponenttypeid` = 4). I tokenize speech into sentences using the NLTK Punkt segmenter, which correctly handles common abbreviations in earnings call language (e.g., “*Mr.*,” “*Inc.*,” “*Q3 2023.*,” “*p.m.*”). The raw sentence corpus contains 65,689,787 sentences across all CEOs and calls.

I filter sentences before scoring; a sentence is kept if it satisfies all of the following:

- (i) Length: between 8 and 60 words inclusive;
- (ii) Pronoun: contains at least one of *I*, *my*, *mine*, *myself*, *we*, *our*, *ours*, *ourselves* or the contractions *I’ve*, *I’m*, *I’d*, *I’ll*, *we’ve*, *we’re*, *we’d*. The objective forms *me* and

us are not in the pattern, and neither is *we'll*;

- (iii) Boilerplate: contains none of a fixed list of formulaic phrases (e.g., “*thank you for the question*”, “*good morning*”, “*operator*”);
- (iv) Numbers: fewer than 15% of characters are digits, to exclude statistics-laden passages that report magnitudes rather than attribution.

The filter retains 40,945,467 sentences (62.3% of the raw corpus). The largest cut is the pronoun requirement (32% of sentences).

Restricting to substantive answers. I classify each analyst question into five types and discard the CEO’s answers to two of them: questions requesting modeling inputs or specific numbers (type C4) and questions requesting clarification or handling the mechanics of the call (type C5). The Q&A-based individualism measure used in every main specification is built on answers to the three substantive question types (C1–C3) alone; these two excluded types account for 23.4% of CEO answer sentences (untabulated).¹ This is not the cut already made by filter (iv) above: filter (iv) removes individual digit-dense sentences, whereas the question-type rule removes whole answers whose *subject* is fixed by the questioner even when the sentences themselves contain few digits. In a C4 or C5 answer the CEO recites a guidance figure, confirms a definition, or hands the question to the CFO, so whatever first-person framing appears is dictated by reporting convention rather than by how the CEO attributes authority.

3.4 Anchor Construction

The scoring axis is built from a curated set of 100 *anchor sentences* operationalizing the two poles of the individualism dimension: ten sentences per pole in each of five situations that recur on earnings calls—performance deviation, major transactions, strategic redirection, external challenges, and forward outlook. Each sentence names a decision,

¹The prepared-remarks score $\text{Indiv}^{\text{Prep}}$, the dependent variable in Table 11, Panel A, and a channel control in Table 9, has no associated analyst question and is therefore unaffected by this restriction.

judgment, responsibility, or expectation and assigns it either to a single individual (“*I made the decision on the timing of this initiative*”) or to a collective (“*The team reached this conclusion through a shared process*”), with no evaluative language about the quality or the outcome of the act. Appendix [Appendix D](#) lists all 100.

Contamination control. So that anchors do not encode confounds correlated with individualism, every anchor must pass three neutrality tests:

- (T1) *Emotional neutrality*: removing the pronoun (“I” or “we”) must leave a sentence with no positive or negative valence;
- (T2) *Result neutrality*: the sentence must not imply firm performance (good or bad);
- (T3) *Capability neutrality*: the sentence must not imply that the CEO is smart, experienced, or prescient.

I enforce these tests by maintaining a contamination blacklist of 61 words spanning four categories: positive performance words (*strong, excellent, record, . . .*), negative performance words (*headwind, decline, weakness, . . .*), capability and confidence words (*experience, vision, conviction, . . .*), and process-quality words (*rigorous, disciplined, comprehensive, . . .*). All anchors are re-scanned via word-boundary regex against the blacklist before encoding.

Anchor quality tests. *First*, the mean sentiment-valence asymmetry between the two poles is below 0.05 in four of the five contexts.² *Second*, the contamination blacklist scan returns zero violations after final revisions.

3.5 Computing the Individualism Score

Every filtered CEO sentence s is scored by projecting its embedding onto a single scoring axis $\hat{\mathbf{u}}$, built once and held fixed:

²The forward-outlook context yields an asymmetry of 0.069, above the threshold. Because the poles are the equal-weighted mean of the five context sub-centroids, this context carries one-fifth of each pole vector. I retain it because the axis must span the recurring earnings-call situations in which CEO answers actually occur; no leave-one-context-out re-estimation is reported.

$$\text{Indiv}_s = \cos(\mathbf{e}_s, \hat{\mathbf{u}}), \quad \hat{\mathbf{u}} = \frac{\mathbf{u}}{\|\mathbf{u}\|}, \quad \mathbf{u} = \mathbf{d} - (\mathbf{d}^\top \mathbf{p}) \mathbf{p}, \quad \mathbf{d} = \bar{\mathbf{a}}_{\text{ind}} - \bar{\mathbf{a}}_{\text{col}}, \quad (1)$$

where $\mathbf{e}_s \in \mathbb{R}^{768}$ is the unit-normalized sentence embedding produced by a pre-trained MPNet sentence-transformer model³ (Reimers and Gurevych, 2019); $\bar{\mathbf{a}}_{\text{ind}}$ and $\bar{\mathbf{a}}_{\text{col}}$ are the two pole means; and $\mathbf{p}$ is a unit-norm pronoun direction constructed in the same embedding space as the average of the normalized differences between eight first-person-singular sentences and their first-person-plural rewrites. Each pole mean is the simple average of the unit-normalized embeddings of that pole’s 5×10 anchors, which—because the five contexts contribute ten sentences each—is the equal-weighted mean of the pole’s five context sub-centroids. The pole means are not themselves rescaled to unit length; it is their difference $\mathbf{d}$, after the pronoun component is removed, that is normalized. The contextual anchors are written in the situations that earnings-call answers actually address, so the axis is defined in the same discourse context in which CEO sentences are scored.

The pronoun projection is applied to the *axis*, once, and not to each sentence score: $\hat{\mathbf{u}}$ is orthogonal to $\mathbf{p}$ by construction, so every score in the paper is already free of that direction and no score is residualized after the fact. The step matters because a raw individual-minus-collective similarity difference can track nothing more than first-person *singular* versus first-person *plural* pronoun frequency—the contrast Chatterjee and Hambrick (2007) measure directly as $I/(I+We)$. Orthogonality to it is therefore imposed by construction rather than discovered. The score is signed so that positive values indicate greater similarity to the individual pole and negative values greater similarity to the collective pole.

³Specifically, `sentence-transformers/all-mpnet-base-v2`, available on HuggingFace.

3.6 Aggregation to Firm-Year

At the transcript-record level, I compute means and medians of *Indiv* within each (transcriptid, section) cell, where section indexes prepared remarks versus Q&A responses. The mean over the retained Q&A answer sentences—answers to C1–C3 questions only—serves as the primary record-level observation, consistent with the use of Q&A responses by Gow et al. (2016): the answers are delivered without a prepared script, and so less filtered than the remarks that open the same call.⁴

The firm-year score is then built in two steps, with no intervening CEO-quarter stage. First, for each identified speaker and calendar year I take the simple mean of that speaker’s record-level Q&A scores at the firm, so every record counts equally and a call event enters with the weight of however many versions of it the database published. Second, where a firm-year carries more than one identified speaker—41.97% of firm-years do, well above the CEO turnover rate because the identification is not filtered on confidence—the firm-year score $\text{Indiv}_{i,t}$ is the score of the speaker appearing on the most records that year (ties, 4.5% of firm-years, are broken by sort order) and the other rows are dropped. The retained value is one speaker’s score, not a blend, and not necessarily that of the CEO in office at the fiscal year end. Across the panel this rule discards 36.8% of the 47,978 speaker-firm-year rows and leaves 30,310 firm-years (Table 1, Steps 3 and 5), each calendar-year mean carried into the panel as that firm’s fiscal-year observation. To facilitate coefficient interpretation, I standardize $\text{Indiv}_{i,t}$ to a *z*-score within the estimation sample and refer to the resulting variable as Indiv_z .

⁴ $\text{Indiv}_{\text{Q\&A}}$ is systematically higher than $\text{Indiv}_{\text{prepared}}$: the firm-year paired *t*-statistic on $\text{Indiv}_{\text{Q\&A}} - \text{Indiv}_{\text{prepared}}$ is 229.7 ($N = 29,734$; mean difference +0.0239 in raw score units, pooled $\rho = 0.630$; Table 2, Panel A). The two sections also carry different information and not merely different levels: with the current CEO’s contemporaneous prepared-remarks score also on the right-hand side (0.4291, $t = 35.3$, the largest of Column (6)’s reported coefficients), a firm’s prepared-remarks history adds no *detectable* increment to its Q&A score (0.0212, $t = 1.4$) while its Q&A history does (0.2213, $t = 15.7$), whereas in the prepared-remarks equation the ranking reverses, 0.3121 ($t = 19.5$) against 0.0148 ($t = 1.0$) (Table 11, Columns (6) and (3)). Entered alone, each term that goes insignificant is strong—0.1126 ($t = 7.6$, Column (5)) in the Q&A equation and 0.1407 ($t = 9.6$, Column (1)) in the prepared-remarks equation—while the own term barely moves (0.2290 to 0.2213, Columns (4) and (6)), so this is a conditioning result and not a difference in raw persistence, and an imprecise coefficient is not a zero. Read adversely, that 0.2213 is itself the confound: the headline Q&A measure inherits about 0.22 SD of its prior CEOs’ individualism per SD, which Table 9 addresses. The sample is the 17,614 firm-years with an observed prior CEO, not transition years alone.

3.7 Construct Validation

I validate the measure on four dimensions:

- *Internal consistency.* The Pearson correlation between Indiv_z measured in odd-numbered calendar quarters and Indiv_z measured in even-numbered calendar quarters of the same CEO-year is $r = 0.635$ ($N = 30,667$ CEO-years), and $r = 0.809$ in the data-sufficient subsample of CEO-years with four-quarter coverage and at least 600 scored Q&A sentences ($N = 9,279$; Table 2, Panel A). Across-year stability of CEO-level Indiv_z is $r = 0.642$ ($N = 30,339$; 0.834 in the same subsample, $N = 8,763$); the full-sample figures are lower because many CEO-years rest on few scored sentences. The estimation sample does not impose that restriction, so the applicable figures are the full-sample split-half and across-year correlations, 0.635–0.642, not 0.809. Classical measurement error of that size attenuates the individualism interaction estimated below toward zero, making the reported estimate a lower bound.
- *Cross-temporal stability.* Year-by-year mean Indiv_z varies within a range of 0.225 standard deviations over the 17-year sample window (Table 2, Panel A). A range does not by itself separate secular drift from year-to-year noise: a monotone rise of 0.225 SD produces the same statistic. The year-fixed-effect R^2 that would separate them is not reported.
- *Relation to speech volume.* The correlation between Indiv_z and the number of scored sentences is -0.054 (Table 2, Panel A).
- *Discriminant validity vs. Chatterjee–Hambrick.* Indiv_z correlates with the Chatterjee and Hambrick (2007) first-person pronoun ratio at Pearson $\rho = +0.177$ (Spearman $+0.179$) at the CEO-quarter level ($N = 122,721$; Table 2, Panel B). Because the pronoun direction is projected out by construction, that figure is evidence that the pronoun projection succeeded against an externally constructed benchmark, not independent evidence of distinctness; it is positive rather than

zero because the Chatterjee–Hambrick measure is a word-count ratio, not the embedding-space pronoun direction that was removed. The substantive test is the horse race in Table 4, Column (5): entered together, the Q&A-based interaction is if anything slightly larger at +0.0747 ($t = 4.02$) while the pronoun-ratio interaction is insignificant at -0.0110 ($t = -0.56$).

3.8 CEO trait or firm culture? A mover-based decomposition

Following [Bertrand and Schoar \(2003\)](#), I ask whether Indiv_z travels with the CEO across firms rather than capturing stable firm-level communication culture. Among the 164 CEOs in the Equilar CEO panel observed for at least two years at each of two firms, the correlation between a CEO’s mean score at one firm and her mean score at the other is $\rho = +0.672$ ($p < 10^{-22}$). In a variance decomposition of Indiv_z on year, firm, and CEO fixed effects, the CEO-unique increment is 10.6% of the variance against 0.76% for the firm-unique increment, the inverse of the ranking a pure firm-culture interpretation predicts. [Appendix B](#) gives the panel construction, the F-tests, the replication on the narrower ExecuComp universe, and the limitations of the mover sample (Table 12).

4 Empirical Analysis

4.1 Pass-Through of Performance Shocks to Employees

My dependent variable is the log change in employment from year $t - 1$ to year t ,

$$\Delta \log(\text{Emp})_{i,t} = \log(\text{EMP}_{i,t}) - \log(\text{EMP}_{i,t-1}),$$

winsorized at the 1%/99% percentiles to mitigate the influence of mergers, divestitures, and reporting irregularities. My key explanatory variable, $\text{Indiv}_{i,z}$, is the firm-year CEO individualism score described in Section 3: it is built from the CEO’s answers to substantive analyst questions, excluding answers to numerical-modeling and procedural-clarification questions and excluding prepared remarks entirely. Those answers are

delivered without a prepared script, and so are less filtered than the remarks that open the same call. The score is standardized within the estimation sample (mean zero, unit standard deviation).

To examine the differential employment response across CEO types, I estimate the following regression:

$$\begin{aligned}\Delta \log(\text{Emp})_{i,t} = & \beta_1 \Delta \text{ROA}_{i,t} + \beta_2 \Delta \text{ROA}_{i,t} \times \text{Indiv}_{i,z} + \beta_3 \text{Indiv}_{i,z} \\ & + \gamma' \text{Controls}_{i,t-1} + \alpha_i + \delta_k + \lambda_t + \epsilon_{i,t},\end{aligned}$$

where $\Delta \text{ROA}_{i,t}$ is the year-over-year change in return on assets (NI/AT), $\text{Controls}_{i,t-1}$ are lagged firm- and CEO-level characteristics including $\log(\text{AT})$, ROA_{t-1} , leverage, cash holdings, sales growth, labor intensity, Tobin's Q, and the lagged labor share, plus contemporaneous CEO tenure, equity-pay ratio, and ownership percentage; α_i , δ_k , and λ_t are firm, Fama–French 48 industry, and year fixed effects, respectively; all variable definitions appear in Table 13. Standard errors are clustered at the firm level. The coefficient of interest is β_2 ; H1 predicts $\beta_2 > 0$.

Table 4 presents the results. Columns (1)–(4) are estimated on a uniform sample of 13,376 firm-years, so that coefficient differences across columns reflect the specification—the industry fixed effects added in Column (2) and the controls added in Columns (3)–(4)—and not changes in the set of firm-years; Column (5) loses twelve observations for which the pronoun-ratio benchmark is unavailable. The panel throughout requires CRSP–Compustat coverage (Table 1, Step 8). β_2 is positive and significant at the 1% level in every column. In my most demanding specification (Column 4), the point estimate is 0.0730, implying that a one-standard-deviation increase in CEO individualism raises the pass-through of a one-unit shock in ΔROA to employment by approximately 7.3 percentage points. For an adverse shock of $\Delta \text{ROA} = -0.05$ and a CEO at the mean of the individualism distribution ($\text{Indiv}_{i,z} = 0$), the implied employment contraction is $\hat{\beta}_1 \times 0.05 \approx 0.38\%$; for a CEO one standard deviation above the mean ($\text{Indiv}_{i,z} = 1$), the implied contraction is $(\hat{\beta}_1 + \hat{\beta}_2) \times 0.05 \approx 0.74\%$ —roughly twice the

contraction at the average CEO. Both are pooled magnitudes, recomputed on the adverse segment in Section 4.2.

The coefficient β_2 remains stable as I add controls, declining only from 0.0753 in Column (1) to 0.0730 in Column (4)—a change of approximately 3%; firm and year fixed effects are already present in Column (1). This stability indicates that the result is not driven by the observable firm- and CEO-level characteristics I control for; formal Oster (2019) bounds are reported in the notes to Table 4 ($\delta^* = 23.8$; $\beta^*(\delta=1) = +0.072$, under that table’s $R_{\max} = 1.3 R_{\text{long}}^2$ convention). β_1 is not: under those same controls it falls 36%, from 0.1183 ($p < 0.01$) to 0.0755 ($p < 0.05$), as Within R^2 doubles. The level effect β_3 is negative and significant in every column (-0.0099 , $t = -4.07$ in Column (4)): individualistic CEOs post structurally slower headcount growth, about a percentage point per standard deviation. Because $\text{Indiv}_{i,z}$ is mean-zero here, β_3 shifts the intercept and not the slope, leaving the pooled magnitudes above alone. But it is not what the scarcity-revealed channel predicts: that channel is a claim about the interaction, and a disposition difference visible outside downturns is one scarcity did not have to reveal.

Column (5) of Table 4 provides a horse-race against the standardized Chatterjee and Hambrick (2007) first-person pronoun ratio $\text{CH}_{i,z}$ to confirm that $\text{Indiv}_{i,z}$ is not a relabeling of an established narcissism proxy. With both $\text{Indiv}_{i,z} \times \Delta\text{ROA}$ and $\text{CH}_{i,z} \times \Delta\text{ROA}$ entered jointly, the Indiv interaction *strengthens* slightly from 0.0730 to 0.0747 ($t = 4.02$), while both CH terms are insignificant and the interaction is wrong-signed ($\beta'_2 = -0.0110$, $t = -0.56$; $\beta'_3 = +0.0011$, $t = 0.54$). The two measures share only 3.1% of their variance at the CEO-quarter level (Table 2, Panel B), a low overlap the pronoun projection largely guarantees, which is why the horse race and not the correlation is the substantive test.

4.2 Asymmetric Shocks: Testing H2

H2 predicts that individualism raises pass-through *asymmetrically*. A scarcity-revealed weighting is visible only when adverse performance forces a binding tradeoff; a symmetric effect would instead indicate a broad stylistic difference in labor management. I

therefore decompose $\Delta ROA_{i,t}$ into its non-positive and non-negative parts:

$$\Delta ROA_{i,t}^- = \min(\Delta ROA_{i,t}, 0), \quad \Delta ROA_{i,t}^+ = \max(\Delta ROA_{i,t}, 0),$$

and estimate the regression with both segments and their respective interactions with $\text{Indiv}_{i,z}$. Table 5 reports the result; its Column (1) reproduces the benchmark from Table 4, Column (4), for direct comparison. The interaction with $\Delta ROA_{i,t}^-$ is positive and significant (Column (2): $\beta = +0.1164$, $t = 3.55$, $p < 0.01$), while the interaction with $\Delta ROA_{i,t}^+$ is positive but much smaller and not statistically significant ($\beta = +0.0328$, $t = 1.16$). A Wald test rejects equality across the two regimes at the 10% level (diff = +0.084, $t_{\text{diff}} = 1.71$, $p = 0.086$). This is the power caveat of Section 2: the upside interaction's implied 95% confidence interval spans zero but also contains the benchmark estimate of 0.0730, so the data locate the differential in downturns without being able to rule out an upside differential as large as the benchmark.

Figure 1 puts those two coefficients on a signed scale, evaluated at the same ± 0.05 shock used above. Because ΔROA^- is non-positive, the positive β_2^- becomes a *downward*-sloping line once the shock is evaluated at a realization: the implied contraction steepens by 0.58 percentage points per standard deviation of $\text{Indiv}_{i,z}$, against 0.16 points on the favourable side. That slope contrast is the asymmetry claim in one picture. Two things the table does not make visible are worth saying. The favourable line lies entirely *below* zero, because the ΔROA^+ main effect is negative: it describes how fast employment falls in good years, not how fast it grows. And the two lines cross about one standard deviation below the mean, which is a difference in levels rather than a reversal of the gradient.

The decomposed main effects matter on their own: ΔROA^- carries +0.2280 ($t = 5.78$) and ΔROA^+ -0.1340 ($t = -2.74$), so pass-through is a downturn phenomenon before CEO type enters and the average firm cuts, rather than adds, labor as performance improves. Read off that segment, the -0.05 shock of Section 4.1 implies a contraction of $0.2280 \times 0.05 \approx 1.14\%$ at the mean CEO and 1.72% at +1 SD—three times the pooled figure there and 2.3 times at +1 SD, where individualism adds about half again rather than double.

Selection on unobservables is not a threat to the downside interaction in particular: because adding controls *strengthens* it, the [Oster \(2019\)](#) bias-adjusted estimate at equal selection rises to $\beta^*(\delta=1) = +0.128$ (notes to Table 4), so no positive degree of proportional selection can overturn it.

4.3 Identification from Exogenous CEO Turnover

If the gradient reflects the CEO rather than a firm characteristic that selects both the CEO and the labor policy, it should survive where the board could not time the departure, though it still chose the successor. Table 6 restricts the panel to firms whose CEO left through death or illness: the [Gentry et al. \(2021\)](#) turnover database, departure codes 1–2, FY2008–2018 (49 firms, 444 firm-years), and an extended set through fiscal 2023 (66 firms, 604 firm-years) combining the authors’ 2023 release with fiscal-2023 successions I hand-coded from an 8-K census of ExecuComp turnover firms. The interaction stays positive and significant on both—+0.2363 ($t = 3.62$) and +0.2027 ($t = 3.26$) for contemporaneous shocks, +0.3177 and +0.3140 for downside shocks, the latter two each with a restricted wild-cluster bootstrap $p = 0.011$, run on the downside interaction alone as the relevant inference with 49 and 66 clusters. Given the sample size these magnitudes show that the effect survives, not that it is larger than in the full panel, and they rest on few events: leave-one-event-firm-out keeps $t \in [2.62, 4.64]$ across all 66 samples, but dropping the two most influential event firms together—both oil-and-gas producers—attenuates the Column (4) downside estimate to +0.155 ($t = 1.43$), so the design is concentrated in, though not driven solely by, energy successions. Excluding FY2020–2021 instead raises that estimate to +0.427 ($t = 3.09$).

4.4 Identification from Retirement-Driven Turnover

The death-and-illness design is credible but small. Table 7 uses a larger quasi-exogenous margin: firms with at least one succession in which the outgoing CEO was aged 60 or older (783 firms, 7,948 firm-years), so that the departure clock is set by age rather than by performance or strategy (cf. [Fee et al., 2013](#)). The contemporaneous interaction is

+0.0746 ($t = 2.90$) and the downside interaction is +0.1161 ($t = 2.62$), the latter with a restricted wild-cluster bootstrap $p = 0.007$. Both are close to the corresponding full-panel estimates in Tables 4 and 5, though this downside column enters ΔROA^- alone, re-estimating that gradient rather than re-testing the asymmetry. The average-CEO term is significant here: against $\beta_1 = +0.0799$ ($t = 2.01$) the +0.0746 interaction again roughly doubles pass-through. But 7,948 firm-years is 59% of the baseline 13,376: not an independent replication. Leave-one-out and drop-two statistics are not reported for this design.

4.5 Where the Burden Lands: Separation Composition

Employment counts do not reveal whether the adjustment falls on workers who are pushed out or on workers who would have left anyway. Table 8 uses Revelio Labs position records to split each firm-year's separations into *rough* (no subsequent position beginning within 90 days, proxying for involuntary) and *smooth*, and estimates the asymmetric specification on the rough share ($N = 13,146$; 1,517 firms—the main panel less the 230 firm-years with no Revelio separation records). Two caveats attach to that outcome: a 90-day gap need not be involuntary, and Revelio coverage is tilted toward positions visible in online professional records. In downturns, as H3 predicts, the rough share rises more steeply for firms run by individualistic CEOs ($\beta^- = -0.0612$, $t = -4.61$, on the weakly negative ΔROA^-), and a Wald test rejects linearity in ΔROA ($p < 0.001$). The upside gradient is not a clean null: at +0.0260 ($t = 1.95$) it points the same way, at less than half the downside magnitude and only marginal significance; no test compares the two magnitudes. The main effects give -0.0612 its scale: the average firm's own downside response is -0.1188 ($t = -8.00$) against an upside -0.0166 ($t = -0.98$), so the interaction is 52% of it. The level term is a precise zero (-0.0006 , $t = -0.65$) against the -0.0099 headcount-growth effect in Table 4, Column (4): the difference is shock-contingent rather than a standing separation regime. Column (2) rescales the outcome to $2 \times n_R/n_S - 1$ and is by construction twice Column (1) with identical t -statistics.

4.6 Additional Robustness

Excluding the COVID-19 period. Re-estimating the benchmark regression after excluding firm-years 2020–2021 ($N = 11,607$) yields $\beta_2 = +0.0929$ ($t = 4.55$, $p < 0.01$; Table 10, Column (1)). The interaction is *stronger* than in the full sample, so the result is not pandemic-specific. The average-CEO term does not survive it: β_1 is 0.0427 ($t = 1.30$) there, so outside that window the data identify heterogeneity across CEOs rather than the average response.

Firm-specific cyclical-sensitivity slopes. A remaining concern is that β_2 could reflect endogenous matching between individualistic CEOs and firms whose employment is intrinsically more sensitive to performance shocks. I therefore absorb, in addition to the firm fixed effect, a firm-specific slope on ΔROA ($\text{Firm} \times c.\Delta\text{ROA}$), so that β_2 is identified *only* from within-firm changes in CEO individualism that move the firm’s ΔROA –employment slope. I restrict the sample to firms with at least five firm-years, a discretionary cut that keeps the firm-specific slopes from being fitted on very short panels; sensitivity to that threshold is not reported. Under this saturated specification (Column (2) of Table 10), the point estimate is 0.0813 ($t = 2.21$) on the 12,440 firm-years that survive the restriction, against 13,376 in the benchmark and with roughly twice its standard error. Its survival, not its size, is what endogenous matching would not predict.

Time-varying firm culture: direct channel controls. The saturated specification above absorbs *time-invariant* matching between CEO type and firm cyclicity, but a slow-moving, time-varying firm culture could in principle jointly shape who is hired, how the firm communicates, and how it adjusts labor. For this confound to operate, it must enter the measure through an observable channel, and Table 11 identifies the two candidates: the firm’s scripted communication and its hiring continuity (the individualism of strictly prior CEOs at the same firm). Table 9 re-estimates the main specification controlling directly for both channels. Columns (1)–(2) open the scripted channel alone: on the 13,156 firm-years with a prepared-statement score it moves β_2 from +0.0819 ($t = 4.23$) to +0.0619 ($t = 2.26$) while its own interaction is insignificant (+0.0360, $t = 1.22$) and its own level term is significant (−0.0117, $t = -4.22$). Columns (3)–(5) also require

an observed prior CEO (11,795), so neither half’s benchmark is the 0.0730 of Table 4, Column (4) (13,376). The prior-CEO history interaction—the empirical carrier of “the firm keeps hiring similar people”—is insignificant in Column (4) (+0.0305, $t = 1.19$), though it attenuates β_2 there by about a fifth (0.0818 $\rightarrow$ 0.0653); with all channels entered jointly in Column (5), the Q&A-based coefficient attenuates by about a quarter but remains significant at $\beta_2 = +0.0594$ ($t = 1.99$). The pass-through effect thus survives conditioning on both channels. It is not cleanly separated from them: in Column (5) the channel interactions are of the same order as β_2 itself (+0.0463, $t = 1.54$, and -0.0546 , $t = -1.32$, for hiring continuity; +0.0437, $t = 1.19$, for scripted communication), and no test in the paper distinguishes any of them from the Q&A coefficient.

The gradient is present in the full panel, larger in downturns than in upturns—though the upside interaction is positive but imprecise and equality of the two interactions is rejected only at the 10% level—robust to the saturated and culture-channel specifications, and holds in both succession designs and in the composition of separations.

5 Conclusion

This paper documents that CEO individualism systematically tilts the burden of adverse performance shocks onto employees. Shareholders are correspondingly insulated, though only the employee margin is measured directly. Using a transformer-based text measure built from CEOs’ answers to substantive analyst questions on quarterly earnings calls—answers delivered without a prepared script, and so less filtered than the remarks that open the same call; answers to purely numerical modeling questions and to procedural clarifications are excluded—I find that a one-standard-deviation increase in CEO individualism raises the pass-through of a one-unit Δ ROA shock to employment by approximately 7.3 percentage points ($\beta_2 = 0.0730$)—roughly doubling the 7.6-point pass-through of a CEO at the sample mean of individualism ($\beta_1 = 0.0755$; Table 4, Column (4)). The differential is concentrated in downturns: individualistic CEOs cut workforces aggressively when performance falls, while in good times it is positive but

imprecisely estimated—though equality of the two interactions is rejected only at the 10% level (Table 5). This asymmetric pattern is consistent with the *scarcity-revealed weighting* channel. A parallel margin in individual position records shows the same downside concentration in the composition of separations, with individualistic CEOs’ downturn separations shifting toward abrupt exits, though the upside gradient there is not a clean null (+0.0260, $t = 1.95$; Table 8). The employment result survives robustness checks including exclusion of the COVID-19 period, which raises the estimate to 0.093 (Table 10, Column (1)), and a horse race against the Chatterjee and Hambrick (2007) pronoun ratio.

The paper makes three contributions. First, I introduce a transformer-based text measure of CEO disposition built from per-sentence semantic similarity to 100 validated anchor sentences—split-half stable on the data-sufficient subsample ($\rho = 0.81$; 0.64 in the full sample, Table 2), built on an axis from which the pronoun direction is projected out, weakly correlated with pronoun frequency as a result ($\rho = +0.18$), and, unlike the pronoun-based narcissism proxy, carrying explanatory power for pass-through (Table 4, Column (5)), and travelling with the CEO across firms more than with the employer (Table 12)—extending text-based measurement in finance (Loughran and McDonald, 2011; Hassan et al., 2019) from sentiment dictionaries to disposition. Second, I extend the CEO-style literature (Bertrand and Schoar, 2003; Custódio et al., 2013; Graham et al., 2013) into the domain of risk allocation between firm constituencies. The labor-related operating leverage studied in that literature (Donangelo, 2014; Donangelo et al., 2019) is a natural place for a managerial-heterogeneity micro-foundation, but establishing one would require a labor-cost outcome, and only the employment margin is measured here. Third, I bring two identification strategies to bear: a saturated specification absorbing firm-specific cyclical slopes (Table 10, Column (2)), which addresses time-invariant CEO–firm matching, and death-, health-, and retirement-driven successions (Tables 6 and 7), which remove the firm’s ability to time the succession though not the board’s choice of successor.

Several limitations bound the conclusions. The text measure cannot fully rule out

strategic communication. The designs above absorb time-invariant matching and set succession timing by health or age rather than by firm performance or labor policy, but they do not absorb every time-varying confounder that co-moves with CEO disposition—although observable firm-culture channels attenuate the CEO interaction by about a fifth to a bit over a quarter across Columns (4) and (5) and leave it significant (Table 9), and Oster (2019) bounds imply unobserved selection would need to be more than twenty times as important as the full control set to overturn the estimate. The death- and health-succession evidence rests on only 49 and 66 event firms, with the extended-set magnitude concentrated in, though not driven solely by, energy successions; the retirement design screens on age rather than on a coded departure reason, and its 7,948 firm-years are a subset of the baseline 13,376 rather than an independent sample. The separation-composition margin rests on a 90-day gap proxy and on Revelio coverage tilted toward positions visible in online professional records. The welfare interpretation rests on the assumption that downturn layoffs impose net costs on employees, and the objective-weighting reading is one of several theoretically consistent interpretations—competing channels include CEO risk tolerance, pay-for-performance incentives, and differential information processing, which the present design cannot separately identify. The scarcity-revealed channel speaks to the interaction, not the level, where individualism enters negatively ($\beta_3 = -0.0099$, $t = -4.07$; Table 4, Column (4)). The documented heterogeneity carries no normative ranking of individualism versus collectivism; it is a behavioral regularity, not a value judgment.

Three extensions are natural: whether the effect is muted where external stakeholder protections are stronger (e.g. union density), which would test the scarcity-revealed weighting channel by varying the constraints on how CEOs allocate adverse shocks; whether CEO compensation contracts symmetrically with the workforce; and whether aggressive labor adjustment improves future performance or imposes costs through morale, turnover, and lost human capital.

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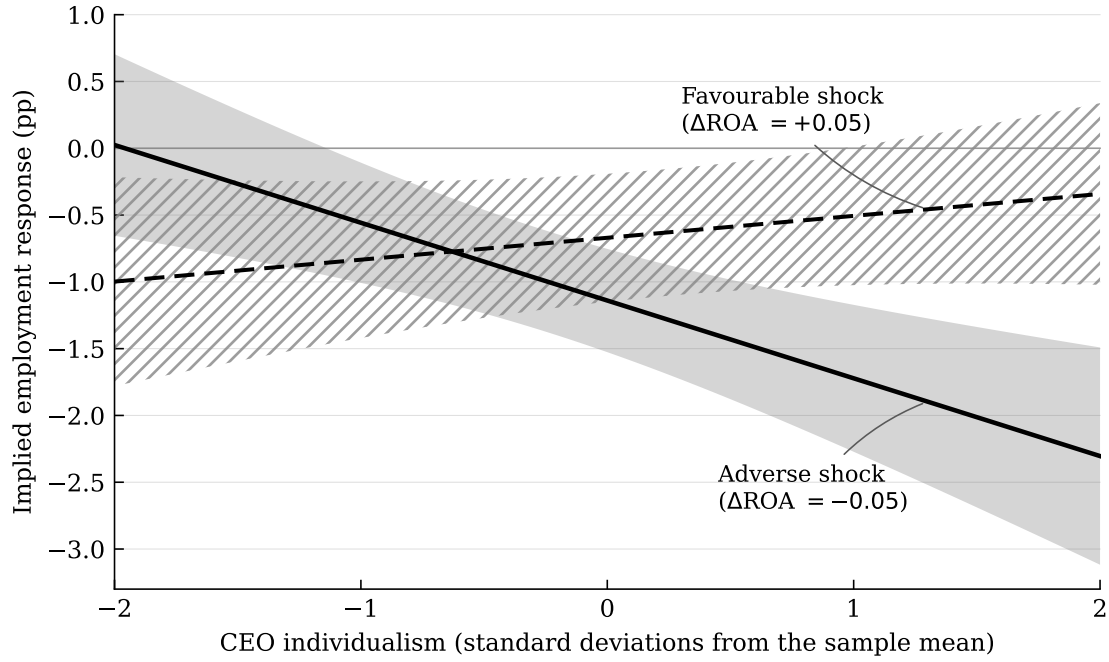


Figure 1: Implied Employment Response to a Performance Shock, by CEO Individualism

Notes. Implied $\Delta \log(\text{Emp})_{i,t}$, in percentage points, from the asymmetric specification of Table 5, Column (2), estimated on the 13,376 firm-years of Table 4, Column (4). Each line is $(\hat{\beta}_1 + \hat{\beta}_2 \text{Indiv}_{i,z}) \times s$ for a shock of size s : the solid line sets $s = -0.05$ and uses the non-positive segment ΔROA^- , the dashed line sets $s = +0.05$ and uses the non-negative segment ΔROA^+ . Shaded and hatched areas are 95% intervals built from the firm-clustered covariance of the two coefficients in each segment, with the same small-sample scaling used for the t -statistics in the tables. The horizontal axis spans two standard deviations either side of the sample mean of the measure, which is standardized within this sample. The figure carries the paper's asymmetry claim as a difference in *slope*: the adverse line falls steeply in individualism, the favourable line is close to flat. Two features should not be over-read. The favourable line lies below zero throughout, because the main effect on ΔROA^+ is itself negative (-0.1340 , $t = -2.74$): the average firm in this panel does not add employment as performance improves, so the dashed line describes how fast employment falls in good years, not how fast it grows. And the two lines cross near one standard deviation below the mean, which reflects those level effects rather than a reversal of the individualism gradient. Equality of the two gradients is rejected only at the 10% level ($p = 0.086$; Table 5).

Table 1: Sample Construction

Notes. Sample construction from raw earnings-call transcripts (Step 1) to the main estimation sample (Step 9) used in Table 4, Column (4), which spans fiscal years 2010–2023. The count falls most at Steps 2–3, which apply the substantive-question (C1–C3) restriction of Section 3 and change the unit of observation from the transcript record to the CEO-quarter (Step 2) and then to the speaker-firm-year (Step 3), and at Steps 5–9 (the collapse to one speaker per firm-year, then the Compustat, industry-exclusion, CRSP-coverage and control-availability filters). Step 5 is not a coverage filter: it discards the rows of every identified speaker other than the retained one, so nearly all of the fall from Step 3 to Step 6 (17,668 of the 17,920 rows lost) is that collapse rather than the Compustat merge. The Step 2 CEO-quarter cells are the unit of the pronoun-ratio comparison in Table 2, Panel B; they are not an intermediate stage in the construction of the firm-year score, which aggregates records directly to the speaker-firm-year level (Section 3). All counts are exact.

Step	Filter applied	Unit of observation	<i>N</i>
<i>Panel A: Text-based individualism construction</i>			
1	Earnings-call transcript records for ExecuComp firms, fiscal years 2008–2024	transcript record	376,002
2	Scored transcript records—those with at least one C1–C3 CEO answer sentence—grouped by calendar quarter and identified CEO speaker, matched by name to the ExecuComp CEO of record (<code>ceoann</code> = CEO); no identification-confidence filter is applied	CEO-quarter	122,721
3	Mean of each identified speaker’s record-level scores within a firm and calendar year (every record weighted equally)	speaker-firm-year	47,978
<i>Panel B: Construct validation sample (Table 2)</i>			
4	Restrict to speaker-years with both odd (Q_1, Q_3) and even (Q_2, Q_4) quarter coverage in the same calendar year	speaker-firm-year	30,667
<i>Panel C: Main estimation sample (Table 4)</i>			
5	Retain, among the Step 3 rows, the score of the speaker appearing on the most records in each firm-year (ties broken by sort order); other rows are dropped	firm-year	30,310
6	Merge to Compustat firm fundamentals (<code>at</code> , <code>sale</code> , <code>emp</code> , <code>ppent</code> , ...)	firm-year	30,058
7	Exclude financial firms and utilities (Fama–French 48 industry classification)	firm-year	21,598
8	Require CRSP–Compustat stock-file coverage (firm-years present in the CRSP–Compustat–ExecuComp master panel; a data-coverage filter, not a CEO-related restriction)	firm-year	19,076
9	Require non-missing lagged controls (size, ROA, leverage, cash, sales growth, labor intensity, Tobin’s Q , ELS) – drops the first firm-year per firm	firm-year	13,376

Table 2: Construct Validation of the Individualism Measure

Notes. Construct-validation tests for the Indiv_z measure, computed on the 364,222 ExecuComp earnings-call transcript records (2008–2024) that retain at least one answer sentence after the C1–C3 question restriction. Indiv_z is the standardized firm-year individualism score. Panel A reports reliability, stability, and the separation between the two sections of the call: the paired t compares each firm-year’s Q&A-based score with the score computed on the prepared-statement section of the same calls, and the two rows beneath it give that comparison its scale—the mean of the paired difference, in raw (unstandardized) score units rather than standard deviations of Indiv_z , and the correlation between the two section scores, both computed on the same 29,734 firm-years. The sentence-count row tests whether the score is mechanically related to how much the CEO speaks. Panel B reports discriminant validity against the Chatterjee–Hambrick (2007) first-person pronoun ratio $I/(I + \text{We})$, measured at the CEO-quarter level. The data-sufficient subsample restricts to CEO-years with four-quarter coverage and ≥ 600 Q&A sentences.

Test	Statistic	N
<i>Panel A: Reliability, stability, and Q&A-vs-prepared separation</i>		
Within-CEO stability (ρ of odd-vs-even quarter Indiv_z in same year)	0.635	30,667
4-quarter, ≥ 600 Q&A sentences subsample	0.809	9,279
Across-year stability (ρ of CEO-year Indiv_z at t vs. $t - 1$)	0.642	30,339
4-quarter, ≥ 600 Q&A sentences subsample	0.834	8,763
Year drift (range of yearly mean Indiv_z , 2008–2024)	0.225	17 years
Q&A vs. prepared remarks (firm-year paired t on $\text{Indiv}_{\text{Q\&A}} - \text{Indiv}_{\text{prep}}$)	$t = 229.7$	29,734
mean difference $\text{Indiv}_{\text{Q\&A}} - \text{Indiv}_{\text{prep}}$ (raw score units)	+0.0239	29,734
ρ between the Q&A and prepared-remark scores	0.630	29,734
Sentence-count independence (ρ between $n_{\text{sentences}}$ and Indiv_z)	−0.054	364,222
<i>Panel B: Discriminant validity vs. Chatterjee–Hambrick (2007) pronoun ratio</i>		
Pearson ρ between Indiv_z and CH_z at CEO-quarter level	+0.177	122,721
Spearman ρ between Indiv_z and CH_z at CEO-quarter level	+0.179	122,721
Implied shared variance ($R^2 = \rho^2$)	0.031	122,721

Table 3: Descriptive Statistics

Notes. The estimation sample of Table 4, Column (4): 13,376 firm-years for 1,562 firms, fiscal years 2010–2023, nonfinancial and nonutility (Table 1, Step 9). $\Delta ROA^- = \min(\Delta ROA, 0)$ and $\Delta ROA^+ = \max(\Delta ROA, 0)$, so the former is non-positive by construction and its mean is negative. $Indiv_{i,z}$ is reported both as scored and as standardized within this sample, which is the form every regression uses. The dependent variable, ΔROA and all controls are winsorized at 1%/99%; $Indiv$ is not. Variable definitions are in Table 13. Panel B reports Pearson correlations on the same sample.

	Mean	SD	p25	Median	p75			
<i>Panel A: Summary statistics</i>								
<i>Dependent variable</i>								
$\Delta \log(\text{Emp})_{i,t}$	0.047	0.161	−0.020	0.032	0.102			
<i>Key regressor</i>								
$\text{Indiv}_{i,z}$ (raw)	0.022	0.019	0.010	0.022	0.035			
$\text{Indiv}_{i,z}$ (standardized)	0.000	1.000	−0.649	0.002	0.670			
<i>Shock</i>								
$\Delta \text{ROA}_{i,t}$	0.001	0.094	−0.023	0.001	0.025			
$\Delta \text{ROA}^-_{i,t}$	−0.026	0.060	−0.023	0.000	0.000			
$\Delta \text{ROA}^+_{i,t}$	0.027	0.062	0.000	0.001	0.025			
<i>Controls</i>								
$\log(\text{AT})_{i,t-1}$	7.802	1.596	6.676	7.693	8.855			
$\text{ROA}_{i,t-1}$	0.041	0.104	0.014	0.050	0.088			
$\text{Leverage}_{i,t-1}$	0.249	0.187	0.092	0.236	0.373			
$\text{Cash}_{i,t-1}$	0.163	0.167	0.041	0.106	0.224			
$\text{Sales growth}_{i,t-1}$	0.097	0.250	−0.014	0.065	0.167			
$\text{Labor intensity}_{i,t-1}$	−4.245	1.537	−4.915	−4.023	−3.298			
$\text{Tobin's } Q_{i,t-1}$	2.164	1.498	1.252	1.674	2.475			
$\text{Labor share (ELS)}_{i,t-1}$	0.267	0.358	0.083	0.168	0.317			
$\text{CEO tenure}_{i,t}$	6.275	3.941	3.000	6.000	9.000			
$\text{Equity pay share}_{i,t}$	0.645	0.281	0.566	0.749	0.840			
$\text{CEO ownership (\%)}_{i,t}$	1.697	4.190	0.113	0.353	1.083			
<i>Panel B: Pearson correlations</i>								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
(1) $\Delta \log(\text{Emp})$	1.00							
(2) Indiv_z	−0.08	1.00						
(3) ΔROA	0.01	0.01	1.00					
(4) $\log(\text{AT})$	−0.12	0.18	−0.03	1.00				
(5) ROA_{t-1}	0.11	−0.01	−0.46	0.18	1.00			
(6) Leverage	−0.11	0.20	0.08	0.38	−0.10	1.00		
(7) Labor int.	0.03	−0.40	0.00	−0.35	0.05	−0.26	1.00	
(8) ELS	−0.05	−0.15	0.01	−0.22	0.02	−0.17	0.51	1.00

Table 4: CEO Individualism and Employment Pass-Through

Notes. DV: $\Delta \log(\text{Emp})$. Sample: ExecuComp firm-years 2010–2023, excluding financials and utilities (FF48). Variable definitions in Table 13. Standard errors clustered by firm; t -statistics in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The interaction β_2 (bolded) is the coefficient of interest. Column (5) augments Column (4) with $\text{CH}_{i,z}$ and $\Delta \text{ROA}_{i,t} \times \text{CH}_{i,z}$ as a horse race. Selection on unobservables (Oster 2019, $R_{\max} = 1.3 R_{\text{long}}^2$): unobserved confounders would need to be $\delta^* = 23.8$ times as important as the full control set to drive β_2 to zero, and the bias-adjusted estimate at equal selection is $\beta^*(\delta=1) = +0.072$; for the downside interaction ($\Delta \text{ROA}^- \times \text{Indiv}_{i,z}$, Table 5), adding controls *strengthens* the coefficient ($\beta^*(\delta=1) = +0.128$), so no positive degree of proportional selection can overturn it.

Dependent variable: $\Delta \log(\text{Emp}_{i,t})$	(1) Base	(2) +Industry FE	(3) +Fin/Labor	(4) +CEO	(5) +CH
$\Delta \text{ROA}_{i,t} \times \text{Indiv}_{i,z} \quad (\beta_2)$	0.0753*** (3.97)	0.0747*** (3.93)	0.0736*** (3.93)	0.0730*** (3.91)	0.0747*** (4.02)
$\Delta \text{ROA}_{i,t} \times \text{CH}_{i,z} \quad (\beta'_2)$	—	—	—	—	-0.0110 (-0.56)
$\Delta \text{ROA}_{i,t} \quad (\beta_1)$	0.1183*** (3.97)	0.1182*** (3.97)	0.0774*** (2.59)	0.0755** (2.53)	0.0765** (2.56)
$\text{Indiv}_{i,z} \quad (\beta_3)$	-0.0091*** (-3.58)	-0.0094*** (-3.73)	-0.0094*** (-3.89)	-0.0099*** (-4.07)	-0.0101*** (-4.03)
$\log(\text{AT})_{t-1}$	-0.0714*** (-11.21)	-0.0723*** (-11.27)	-0.0750*** (-11.17)	-0.0778*** (-11.46)	-0.0780*** (-11.49)
ROA_{t-1}	0.3824*** (10.33)	0.3823*** (10.27)	0.2457*** (6.26)	0.2417*** (6.15)	0.2418*** (6.15)
$\text{CH}_{i,z} \quad (\beta'_3)$	—	—	—	—	0.0011 (0.54)
Leverage_{t-1}	—	—	-0.0595*** (-3.02)	-0.0572*** (-2.92)	-0.0573*** (-2.93)
Cash_{t-1}	—	—	0.2270*** (8.26)	0.2288*** (8.35)	0.2291*** (8.36)
SalesGrowth_{t-1}	—	—	0.0168** (1.98)	0.0168** (1.98)	0.0170** (2.00)
$\text{LaborIntensity}_{t-1}$	—	—	-0.0352*** (-4.84)	-0.0350*** (-4.81)	-0.0351*** (-4.82)
$\text{Tobin } Q_{t-1}$	—	—	0.0231*** (9.87)	0.0223*** (9.60)	0.0223*** (9.60)
ELS_{t-1}	—	—	-0.1969*** (-7.21)	-0.1975*** (-7.26)	-0.1975*** (-7.24)
CEO Tenure	—	—	—	0.0015*** (2.94)	0.0015*** (2.92)
CEO Equity Pay	—	—	—	0.0260*** (3.32)	0.0259*** (3.30)
CEO Ownership	—	—	—	-0.0010 (-1.28)	-0.0010 (-1.25)
Firm FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
FF48 Industry FE	—	Yes	Yes	Yes	Yes
Cluster SE by firm	Yes	Yes	Yes	Yes	Yes
N	13,376	13,376	13,376	13,376	13,364
Within R^2	0.060	0.061	0.116	0.118	0.118

Table 5: Asymmetric Employment Pass-Through: Downturns versus Upturns

Notes. Test of the scarcity-revealed weighting prediction that the individualism differential in pass-through is concentrated in downturns. Column (1) repeats the baseline symmetric specification (Table 4, Column (4)). Column (2) decomposes the performance shock into $\Delta ROA^- = \min(\Delta ROA, 0)$ and $\Delta ROA^+ = \max(\Delta ROA, 0)$ and interacts each part with *Indiv*. The Wald test compares the downside and upside interaction coefficients. Firm- and CEO-level controls match the baseline (suppressed). SEs clustered by firm; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$; coefficients of interest bolded.

Dependent variable: $\Delta \log(\text{Emp}_{i,t})$	(1) Baseline	(2) Asymmetric Shocks
$\Delta ROA_{i,t} \times \text{Indiv}_{i,z} \quad (\beta_2)$	0.0730 *** (3.91)	—
$\Delta ROA_{i,t}^- \times \text{Indiv}_{i,z} \quad (\beta_2^-)$	—	0.1164 *** (3.55)
$\Delta ROA_{i,t}^+ \times \text{Indiv}_{i,z} \quad (\beta_2^+)$	—	0.0328 (1.16)
$\Delta ROA_{i,t} \quad (\beta_1)$	0.0755** (2.53)	—
$\Delta ROA_{i,t}^-$	—	0.2280*** (5.78)
$\Delta ROA_{i,t}^+$	—	-0.1340*** (-2.74)
$\text{Indiv}_{i,z} \quad (\beta_3)$	-0.0099*** (-4.07)	-0.0072*** (-2.86)
<i>Wald test:</i> $\beta_2^- = \beta_2^+$ in Col. (2): diff = +0.084, $t = 1.71$, $p = 0.086$ (rejects equality at the 10% level; pass-through is concentrated in downturns)		
Firm FE	Yes	Yes
Year FE	Yes	Yes
FF48 Industry FE	Yes	Yes
Firm- and CEO-level controls	Yes	Yes
Cluster SE	firm	firm
N	13,376	13,376

Table 6: Identification from Exogenous CEO Turnover: Death- and Health-Driven Successions

Notes. Employment pass-through of CEO individualism identified from successions whose *timing* is exogenous: CEOs who left through *death* or *health/illness*. The board still chooses the successor; what the design removes is the firm's ability to time that choice to coincide with a planned change in labor policy—the reverse-causality channel of concern. Columns (1)–(2) use the [Gentry et al. \(2021\)](#) open-source CEO-turnover database, departure codes 1–2, FY2008–2018 (49 firms, 444 firm-years). Columns (3)–(4) extend it with the authors' 2023 release and fiscal-2023 successions I hand-coded from an 8-K census of ExecuComp turnover firms (66 firms, 604 firm-years). Odd columns use $\Delta ROA_{i,t}$; even columns the downside shock $\Delta ROA_{i,t}^- = \min(\Delta ROA_{i,t}, 0)$. All columns include firm, FF48-industry and year fixed effects and the full control set of Table 4; SEs clustered by firm, t -statistics in parentheses, restricted wild-cluster bootstrap p -values (Rademacher, $B = 1,999$) for the downside interaction—the relevant inference with only 49 and 66 clusters. Column (4) is stable under stress: leave-one-event-firm-out leaves $t \in [2.62, 4.64]$ in all 66 samples, and excluding FY2020–2021 gives $\beta_2^- = +0.427$ ($t = 3.09$); dropping the two most influential event firms together (both oil-and-gas producers) attenuates it to $+0.155$ ($t = 1.43$), so the extended-set estimate is concentrated in—though not driven solely by—energy successions. Given the sample size, the magnitudes should be read as evidence that the effect survives, not that it is larger than in the full panel. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	(1)	(2)	(3)	(4)
	GHQB 2021 release, 2008–18		Extended set, 2008–23	
Dependent variable: $\Delta \log(\text{Emp}_{i,t})$	Contemp.	Downside	Contemp.	Downside
$\Delta ROA_{i,t} \times \text{Indiv}_{i,z} \quad (\beta_2)$	0.2363*** (3.62)	—	0.2027*** (3.26)	—
$\Delta ROA_{i,t}^- \times \text{Indiv}_{i,z} \quad (\beta_2)$	—	0.3177*** (3.45)	—	0.3140*** (4.20)
<i>Wild-cluster bootstrap p</i>	—	0.011	—	0.011
$\Delta ROA_{i,t} \quad (\beta_1)$	0.2254 (1.21)	—	0.1316 (0.77)	—
$\Delta ROA_{i,t}^- \quad (\beta_1)$	—	0.5162*** (2.73)	—	0.4207** (2.26)
$\text{Indiv}_{i,z} \quad (\beta_3)$	−0.0270** (−2.55)	−0.0228** (−2.06)	−0.0189** (−2.53)	−0.0135* (−1.81)
Firm + FF48 + Year FE	Yes	Yes	Yes	Yes
Firm- and CEO-level controls	Yes	Yes	Yes	Yes
Cluster SE	firm	firm	firm	firm
N / Firms	444 / 49	444 / 49	604 / 66	604 / 66
Within R^2	0.173	0.192	0.141	0.157

Table 7: Identification from Retirement-Driven CEO Turnover

Notes. This table complements the death- and health-succession design of Table 6 with a second, larger quasi-exogenous turnover margin: successions whose *timing* is driven by the outgoing CEO's age rather than by performance or strategy (cf. Fee et al., 2013). CEO transitions are identified from the matched-CEO panel; the outgoing CEO's age is taken from ExecuComp in the fiscal year before the transition. The sample retains firms with at least one succession in which the outgoing CEO was aged 60 or older (783 firms, 7,948 firm-years). Because the departure clock is set by age, what the design removes is the firm's ability to time the succession to coincide with a planned change in labor policy. The screen is on the *outgoing* CEO's age and says nothing about who is hired next, so the board's choice of successor is left in place here as it is in Table 6. Column (1) uses $\Delta ROA_{i,t}$; column (2) the downside shock $\Delta ROA_{i,t}^-$. Both columns include firm, FF48-industry and year fixed effects and the full control set of Table 4; SEs clustered by firm. The column (2) estimate carries a restricted wild-cluster bootstrap p -value of 0.007 (Rademacher, $B = 1,999$). Leave-one-out and drop-two stress statistics are not reported for this table. The downside magnitude (+0.12) is close to the full-sample estimate (Table 5), but the two are not independent: the 7,948 firm-years here are a subset of the 13,376 estimated there. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Dependent variable: $\Delta \log(\text{Emp}_{i,t})$	(1)	(2)
	Outgoing CEO age ≥ 60	
	Contemp.	Downside
$\Delta ROA_{i,t} \times \text{Indiv}_{i,z} \quad (\beta_2)$	0.0746*** (2.90)	—
$\Delta ROA_{i,t}^- \times \text{Indiv}_{i,z} \quad (\beta_2)$	—	0.1161*** (2.62)
<i>Wild-cluster bootstrap p</i>	—	0.007
$\Delta ROA_{i,t} \quad (\beta_1)$	0.0799** (2.01)	—
$\Delta ROA_{i,t}^- \quad (\beta_1)$	—	0.1716*** (3.17)
$\text{Indiv}_{i,z} \quad (\beta_3)$	-0.0092*** (-3.17)	-0.0059** (-1.96)
Firm + FF48 + Year FE	Yes	Yes
Firm- and CEO-level controls	Yes	Yes
Cluster SE	firm	firm
N / Firms	7,948 / 783	7,948 / 783
Within R^2	0.107	0.109

Table 8: Separation Composition: Asymmetric Pattern in Individual Position Records

Notes. Asymmetric specification of the form $Y_{i,t} = \alpha + \beta_1^- \Delta ROA_{i,t}^- + \beta_1^+ \Delta ROA_{i,t}^+ + \beta^- (\Delta ROA_{i,t}^- \times \text{Indiv}_{i,z}) + \beta^+ (\Delta ROA_{i,t}^+ \times \text{Indiv}_{i,z}) + \beta_3 \text{Indiv}_{i,z} + \gamma X_{i,t} + \alpha_i + \delta_t + \delta_j + \varepsilon_{i,t}$, mirroring Table 5, Column (2). Outcomes are constructed from Revelio Labs *Positions* records (every employee-firm spell, 2008–2024) linked to Compustat via the *rcid-gvkey* crosswalk in *CompanyMapping*. For each ending position, the separation is classified as *rough* if the same employee has no subsequent position starting within 90 days of the end date (proxying for involuntary separation), and *smooth* otherwise (proxying for voluntary moves). The 90-day rule is a proxy: a gap of that length need not be involuntary, and Revelio coverage is tilted toward positions visible in online professional records. Column (1) uses the rough share n_R/n_S (layoff-like share of total separations); Column (2) uses the rough minus smooth premium $(n_R - n_{Sm})/n_S$, which by construction equals $2 \times n_R/n_S - 1$ and therefore has coefficients exactly $2 \times$ those of Column (1) with identical t -statistics. $\Delta ROA^- = \min(\Delta ROA, 0)$; $\Delta ROA^+ = \max(\Delta ROA, 0)$. $\text{Indiv}_{i,z}$ is the standardized individualism measure of Table 4. Sample as in Table 4 (CRSP-Compustat restricted, fin/utility excluded, 2010–2023), further restricted to firm-years with Revelio separation records. Firm-, FF48-, and Year-fixed effects estimated by within transformation (pyhdfe); full controls suppressed (log AT, lagged ROA, leverage, cash, sales growth, labor intensity, Tobin Q, ELS, CEO tenure, equity pay, ownership). ΔROA , the outcome and the controls are winsorized at 1%/99%; $\text{Indiv}_{i,z}$ is not. SEs clustered by firm; t -statistics in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Dependent variable:	(1) Rough share n_R/n_S	(2) Rough – Smooth $(n_R - n_{Sm})/n_S$
$\Delta ROA_{i,t}^- \times \text{Indiv}_{i,z} \quad (\beta^-)$	–0.0612*** (–4.61)	–0.1225*** (–4.61)
$\Delta ROA_{i,t}^+ \times \text{Indiv}_{i,z} \quad (\beta^+)$	0.0260* (1.95)	0.0520* (1.95)
$\Delta ROA_{i,t}^-$	–0.1188*** (–8.00)	–0.2375*** (–8.00)
$\Delta ROA_{i,t}^+$	–0.0166 (–0.98)	–0.0332 (–0.98)
$\text{Indiv}_{i,z}$	–0.0006 (–0.65)	–0.0012 (–0.65)
$H_0: \beta^- = \beta^+$ Wald p -value	0.000	0.000
Firm- and CEO-level controls	Yes	Yes
Firm FE	Yes	Yes
FF48 Industry FE	Yes	Yes
Year FE	Yes	Yes
Cluster SE	firm	firm
N	13,146	13,146
Firms	1,517	1,517
Within R^2	0.034	0.034

Table 9: Employment Pass-Through: Controlling for Firm-Level Culture Channels

Notes. The main pass-through specification (Table 4, col. 4) re-estimated while controlling directly for the two observable channels through which a slow-moving firm communication culture could enter the measure. $\text{Indiv}_{i,z}^{\text{Prep}}$ is the current CEO's individualism scored on the prepared-statement section of the same calls—the scripted, IR-reviewed channel that Table 11 shows carries firm-level continuity. $\text{Hist}_{i,z}^{\text{QA}}$ and $\text{Hist}_{i,z}^{\text{Prep}}$ are the predetermined firm-history terms of Table 11: call-weighted average individualism of strictly prior CEOs at the same firm, by section. Each channel enters both in level and interacted with $\Delta\text{ROA}_{i,t}$. Columns (1)–(2) use the sample with a prepared-statement score; columns (3)–(5) additionally require an observed prior CEO, with column (3) re-estimating the benchmark on that subsample. All scores standardized within the estimation sample. SEs clustered by firm; t -statistics in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Dependent variable: $\Delta \log(\text{Emp}_{i,t})$ Sample:	(1) Benchmark	(2) +Prep Prep available	(3) Benchmark	(4) +Hist Prior CEO observed	(5) All channels
$\Delta\text{ROA}_{i,t} \times \text{Indiv}_{i,z}^{\text{QA}} \quad (\beta_2)$	0.0819*** (4.23)	0.0619** (2.26)	0.0818*** (4.19)	0.0653*** (2.72)	0.0594** (1.99)
$\Delta\text{ROA}_{i,t} \times \text{Indiv}_{i,z}^{\text{Prep}}$	—	0.0360 (1.22)	—	—	0.0437 (1.19)
$\Delta\text{ROA}_{i,t} \times \text{Hist}_{i,z}^{\text{QA}}$	—	—	—	0.0305 (1.19)	0.0463 (1.54)
$\Delta\text{ROA}_{i,t} \times \text{Hist}_{i,z}^{\text{Prep}}$	—	—	—	—	−0.0546 (−1.32)
$\Delta\text{ROA}_{i,t}$	0.0471* (1.66)	0.0436 (1.51)	0.0404 (1.39)	0.0409 (1.41)	0.0373 (1.27)
$\text{Indiv}_{i,z}^{\text{QA}}$	−0.0096*** (−3.85)	−0.0064** (−2.45)	−0.0109*** (−4.16)	−0.0110*** (−4.15)	−0.0081*** (−2.94)
$\text{Indiv}_{i,z}^{\text{Prep}}$	—	−0.0117*** (−4.22)	—	—	−0.0108*** (−3.68)
$\text{Hist}_{i,z}^{\text{QA}}$	—	—	—	0.0007 (0.15)	−0.0017 (−0.34)
$\text{Hist}_{i,z}^{\text{Prep}}$	—	—	—	—	0.0066 (1.26)
Firm + FF48 + Year FE	Yes	Yes	Yes	Yes	Yes
Firm + labor + CEO controls	Yes	Yes	Yes	Yes	Yes
Cluster SE by firm	Yes	Yes	Yes	Yes	Yes
N	13,156	13,156	11,795	11,795	11,795
Within R^2	0.119	0.121	0.120	0.120	0.123

Appendix A Additional Robustness: COVID and Firm-Specific Cyclical Slopes

Table 10 reports two extensions of the baseline employment specification (Table 4, Column (4)). Column (1) drops fiscal years 2020–2021, so that the interaction is not identified off the pandemic; Column (2) separately absorbs a firm-specific slope on ΔROA . The two are alternative specifications, not nested: Column (2) is estimated on 12,440 firm-years and Column (1) on 11,607.

Table 10: Additional Robustness: COVID Exclusion and Firm-Specific Cyclical-Sensitivity Slopes

Notes. Two robustness extensions of Table 4, Column (4). Column (1) excludes fiscal years 2020–2021 (COVID-19). Column (2) absorbs a firm-specific slope on ΔROA (Firm $\times$ c. ΔROA); β_2 is identified only from within-firm variation in CEO individualism, restricted to firms with ≥ 5 firm-years. The asymmetric decomposition formerly reported alongside these columns appears in Table 5. Firm- and CEO-level controls match the baseline (suppressed). SEs clustered by firm; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$; β_2 bolded.

Dependent variable: $\Delta \log(\text{Emp}_{i,t})$	(1) Drop COVID (2020–2021)	(2) Firm-Slope FE ($n \geq 5$)
$\Delta ROA_{i,t} \times \text{Indiv}_{i,z} \quad (\beta_2)$	0.0929*** (4.55)	0.0813** (2.21)
$\Delta ROA_{i,t} \quad (\beta_1)$	0.0427 (1.30)	—
$\text{Indiv}_{i,z} \quad (\beta_3)$	−0.0094*** (−3.58)	−0.0094*** (−4.00)
Firm FE	Yes	Yes
Firm $\times$ c. ΔROA	—	Yes
Year FE	Yes	Yes
FF48 Industry FE	Yes	Yes
Firm- and CEO-level controls	Yes	Yes
Cluster SE	firm	firm
N	11,607	12,440

Channel purity of the individualism measure

The measure is scored on the CEO's answers in the Q&A section, and the prepared-statement section of the same calls provides a scripted comparison written under the firm's own investor-relations apparatus. Table 11 regresses each section's score on the firm's history in *both* sections, so that firm-level continuity can be attributed to the channel that carries it. Section 3 reports what the two equations show and the sense in which the result is conditional; the mover test below approaches the same CEO-versus-firm question from the opposite direction, holding the CEO fixed and varying the firm.

Table 11: Channel-Pure Firm-Level Persistence of the Individualism Measure

Notes. Does the individualism score reflect the current CEO's disposition, or a slower-moving firm-level signal that survives CEO transitions? Indiv is scored separately on the prepared-statement section ($\text{Indiv}_{i,t,z}^{\text{Prep}}$) and the Q&A section ($\text{Indiv}_{i,t,z}^{\text{QA}}$) of each call and standardized. For each (firm i , current CEO c), the firm-history terms $\text{hist Q\&A}_{i,c}$ and $\text{hist Prep}_{i,c}$ are call-weighted averages of the corresponding score over *strictly prior* CEOs at the same firm, then standardized; firms with no observed prior CEO drop. Panel A (cols. 1–3) regresses $\text{Indiv}_{i,t,z}^{\text{Prep}}$ on $\text{Indiv}_{i,t,z}^{\text{QA}}$ and the history terms; Panel B (cols. 4–6) is the mirror. Columns vary which history terms enter. Controls $X_{i,t}$: $\log(\text{AT})_{t-1}$, Lev_{t-1} , SalesGrowth_{t-1} , ROA_{t-1} , CEO tenure; fixed effects are FF48 industry and calendar year; continuous variables winsorized at 1%/99%. Replacing industry with firm fixed effects preserves the same diagonal (untabulated). SEs clustered by firm; t -statistics in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Dependent variable:	(1)	(2)	(3)	(4)	(5)	(6)
	$\text{Indiv}_{i,t,z}^{\text{Prep}}$ (Panel A)			$\text{Indiv}_{i,t,z}^{\text{QA}}$ (Panel B)		
Firm-history terms entered:	hist Q&A	hist Prep	both	hist Q&A	hist Prep	both
$\text{Indiv}_{i,t,z}^{\text{QA}} \quad (\beta_1)$	0.4493*** (38.1)	0.4014*** (36.5)	0.3979*** (35.9)	—	—	—
$\text{Indiv}_{i,t,z}^{\text{Prep}} \quad (\beta_1)$	—	—	—	0.4361*** (39.0)	0.4558*** (36.4)	0.4291*** (35.3)
$\text{hist Q\&A}_{i,c} \quad (\beta_2)$	0.1407*** (9.6)	—	0.0148 (1.0)	0.2290*** (17.4)	—	0.2213*** (15.7)
$\text{hist Prep}_{i,c} \quad (\beta_3)$	—	0.3180*** (21.9)	0.3121*** (19.5)	—	0.1126*** (7.6)	0.0212 (1.4)
FF48 Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Cluster SE	firm	firm	firm	firm	firm	firm
N	17,614	17,614	17,614	17,614	17,614	17,614
Firms	1,704	1,704	1,704	1,704	1,704	1,704
Within R^2	0.293	0.364	0.364	0.328	0.292	0.328

Appendix B Robustness: CEO-Mover Test of Disposition Portability

The main results in Table 4 implicitly treat $\text{Indiv}_{i,z}$ as a CEO-level disposition. A residual concern is that the measure may instead reflect a stable firm-level communication culture—a possibility that the saturated specification absorbing firm-specific ΔROA slopes (Table 10, Column (2)) addresses only indirectly. This appendix implements the canonical Bertrand and Schoar (2003) “manager fixed effect” identification strategy that directly distinguishes the two explanations: if $\text{Indiv}_{i,z}$ is a CEO-level disposition, the *same* CEO should carry the *same* score across the *different* firms she leads; if it is firm culture, the score should re-set when the CEO changes employer, and firm identity should explain more of its variance than CEO identity does. Throughout this appendix, as in the main tables, $\text{Indiv}_{i,z}$ is scored on the CEO’s unscripted answers in the question-and-answer session and excludes the prepared statement entirely; within the Q&A, answers that merely restate figures for modeling purposes or that handle clarification and procedure are not scored. The two segments are related but not interchangeable. Table 2 rejects equality of their *levels* within firm-years, but on $N = 29,734$ firm-years and against a pooled correlation of 0.630 across those firm-years, so the rejection is not by itself evidence that the segments carry different information. That they carry different information rests instead on Table 11: conditional on the sitting CEO’s own contemporaneous prepared-remarks score and on the firm’s Q&A history under strictly prior CEOs, the firm’s prepared-remarks history carries no detectable additional information about her Q&A score, and symmetrically (Columns (6) and (3)). Entered alone those history terms are 0.1126 ($t = 7.6$) and 0.1407 ($t = 9.6$): the null is a conditioning result, and an imprecise coefficient is not a zero. That table is estimated on every firm-year with an observed prior CEO, not on transition years alone.

Identification strategy and results

I implement three nested tests drawn from the [Bertrand and Schoar \(2003\)](#) “Managing with Style” framework, each stated below with its result; Table 12 reports all three. Column (1) uses the Equilar Insight Data Feed, which follows a CEO across firms through a stable `person_id`; Column (2) uses the ExecuComp `execid`, narrower but well-validated coverage of S&P 1500 firms.

Test 1 (mover correlation). For each CEO who appears at two or more distinct firms with at least two fiscal-year observations at each—so the firm-level mean is not a single call year—I correlate her firm-specific mean $\text{Indiv}_{i,z}$ across her two longest-tenured firms; firm culture predicts $\rho \rightarrow 0$. The estimate is $\rho = +0.672$ ($p < 10^{-22}$, $N = 164$ pairs) in Equilar and $+0.667$ ($p < 10^{-11}$, $N = 82$) in ExecuComp, with Spearman correlations of $+0.694$ and $+0.662$.

Test 2 (CEO fixed-effect F-test). On the sub-panel of firms with ≥ 2 CEOs ($N = 27,058$ firm-years in Equilar, 18,065 in ExecuComp) I estimate four nested fixed-effects models of $\text{Indiv}_{i,z}$: (M1) year only; (M2) year + firm; (M3) year + CEO; (M4) year + firm + CEO. The F -test of M4 against M2 sets to zero the 2,664 CEO dimensions in the Equilar panel that firm identity does not already span, and so asks whether CEO identity adds explanatory power *beyond* firm identity; it gives $F = 2.83$ ($p < 10^{-300}$). The symmetric test of M4 against M3 sets to zero the 216 firm dimensions not spanned by CEO identity and also rejects, at $F = 2.48$ ($p < 10^{-27}$); ExecuComp gives $F = 3.70$ and $F = 2.89$ on 1,696 and 91 dimensions. Only those dimensions are separately identified (a firm whose CEOs appear nowhere else has its effect absorbed by theirs), so the firm side of the test is identified only within the sets of firms that movers connect. Per restriction, then, the CEO component is the larger of the two in both samples, but only by about 14% in Equilar and 28% in ExecuComp—a firm-level component coexists with it.

Test 3 (AKM-style variance decomposition). Following the [Abowd et al. \(1999\)](#) two-way fixed-effects logic adopted by [Bertrand and Schoar \(2003\)](#), I take $\Delta R^2_{\text{CEO}|\text{Firm}} = R^2(\text{M4}) - R^2(\text{M2})$ and $\Delta R^2_{\text{Firm}|\text{CEO}} = R^2(\text{M4}) - R^2(\text{M3})$ as the shares of $\text{Indiv}_{i,z}$ variance uniquely attributable to CEO identity (conditional on firm) and to firm identity

(conditional on CEO). The four models explain 0.9%, 57.4%, 67.3% and 68.1% of that variance in Equilar and 1.0%, 58.3%, 70.2% and 70.7% in ExecuComp, and the increments, computed from unrounded values, are 10.6% against 0.76%, a 14:1 ratio, and 12.3% against 0.52%, or 24:1. Two levels carry a reading of their own: year and firm effects alone already explain 57.4% and 58.3% of $\text{Indiv}_{i,z}$, the strongest firm-side statistic in the paper; and year and CEO effects explain 67.3% and 70.2%, roughly ten and twelve points more on the same panels—though CEO identity contributes far more free dimensions (2,664 against 216 in Equilar), so the comparison carries the same per-parameter qualification the ratio does. The increments themselves are unadjusted rather than per-parameter quantities and no limited-mobility correction is applied to either; the per-parameter comparison is the F -pair.

Interpretation

The three tests deliver a coherent verdict: the CEO-incremental component of $\text{Indiv}_{i,z}$ dominates the firm-incremental component, and on all three margins the ordering runs opposite to what the *pure* firm-culture account set out above would produce. A firm component is nonetheless present—the firm F -test rejects—consistent with two readings: that earnings calls are partially scripted, so a CEO’s disposition is filtered through firm-specific investor-relations practice, or that the CEO adapts in part to the firm’s existing norms. Neither reading disturbs the ordering the three tests establish. That ordering complements what Table 10, Column (2) delivers, along a distinct identification dimension: the saturated specification rules out time-invariant CEO-firm *matching* on intrinsic cyclical sensitivity, while the mover test holds the *CEO* fixed and varies the firm.

Limitations

The mover sub-sample is small—164 of the 5,749 unique CEOs in the Equilar panel (2.9%) and 82 of the 4,407 in ExecuComp (1.9%), because cross-firm transitions are rare and the requirement of ≥ 2 years of calls at each firm filters out short tenures—so

$\rho \approx 0.67$, while clearly distinguishable from zero in both samples ($p < 10^{-11}$), is not tightly bounded. Movers are also a selected cohort: CEOs who survive multiple appointments may differ systematically from those who do not. The variance decomposition is estimated on the full sub-panel of firms with ≥ 2 CEOs rather than on movers alone, a partial check against that concern. Finally, mover identification depends on accurate cross-firm CEO linkage; two separately maintained tracking sources (ExecuComp `execid` and Equilar `person_id`) deliver the same ordering, which indicates that linkage error idiosyncratic to either vendor's identifier is not driving the finding.

Table 12: CEO-Mover Test of Disposition Portability

Notes. **Bertrand and Schoar (2003)** “Managing with Style” mover test of whether $\text{Indiv}_{i,z}$ is a CEO-level disposition or a firm-level communication culture. Panel A reports the Pearson and Spearman cross-firm correlations of a CEO’s mean $\text{Indiv}_{i,z}$ at her two longest-tenured firms, restricted to CEOs with ≥ 2 fiscal-year observations at each of ≥ 2 distinct firms (mover pairs). Panel B reports joint F -tests for CEO and firm fixed effects in nested fixed-effects models of $\text{Indiv}_{i,z}$ on the sub-panel of firms with ≥ 2 CEOs. Panel C reports the AKM-style variance decomposition: the incremental contribution of CEO identity to the R^2 of $\text{Indiv}_{i,z}$ beyond year and firm fixed effects ($\Delta R^2_{\text{CEO}|\text{Firm}}$), and the symmetric contribution of firm identity beyond year and CEO fixed effects ($\Delta R^2_{\text{Firm}|\text{CEO}}$). Column (1) uses the Equilar Insight Data Feed CEO panel (management_role_ranked = CEO); Column (2) uses the ExecuComp (gvkey, excecld) CEO-appointment record. Indiv scores are constructed at the firm-year level from the within-sample standardized $\text{Indiv}_{i,z}$ measure used in Table 4; sample sizes reflect firm-year observations in the relevant subsample. p -values reported in scientific notation where the magnitude exceeds machine precision. Increments and ratios are computed from unrounded values and need not reproduce exactly from the rounded R^2 levels. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

CEO-tracking source	(1) Equilar (primary)	(2) ExecuComp (validation)
<i>Panel A. Mover correlation: $\rho(\text{Indiv}_A, \text{Indiv}_B)$ across firms for same CEO</i>		
Pearson correlation ρ	+0.672***	+0.667***
p -value	$<10^{-22}$	$<10^{-11}$
Spearman correlation ρ_s	+0.694***	+0.662***
Mover pairs	164	82
<i>Panel B. F-tests: marginal contribution of CEO vs Firm fixed effects</i>		
CEO FE Firm FE	$F = 2.83^{***}$	$F = 3.70^{***}$
(p -value)	$<10^{-300}$	$<10^{-300}$
Firm FE CEO FE	$F = 2.48^{***}$	$F = 2.89^{***}$
(p -value)	$<10^{-27}$	$<10^{-17}$
<i>Panel C. Variance decomposition of $\text{Indiv}_{i,z}$ (AKM-style)</i>		
R^2 (Year FE)	0.9%	1.0%
R^2 (Year + Firm FE)	57.4%	58.3%
R^2 (Year + CEO FE)	67.3%	70.2%
R^2 (Year + Firm + CEO FE)	68.1%	70.7%
$\Delta R^2_{\text{CEO} \text{Firm}}$	10.6%	12.3%
$\Delta R^2_{\text{Firm} \text{CEO}}$	0.76%	0.52%
Ratio $\Delta R^2_{\text{CEO} \text{Firm}}/\Delta R^2_{\text{Firm} \text{CEO}}$	14:1	24:1
Total firm-year obs in CEO panel	33,066	23,920
Unique firms	2,548	2,269
Unique CEOs	5,749	4,407
Firms with ≥ 2 CEOs (decomposition subsample)	1,785	1,433
Decomposition N (firm-years)	27,058	18,065

Appendix C Variable Definitions

Table 13: Variable Definitions

Notes. Unless stated otherwise, definitions refer to the main estimation sample: ExecuComp firm-years 2010–2023 (construction and window in Table 1), excluding financial firms and utilities under the Fama–French 48 industry classification. Ratio and difference variables are winsorized at 1%/99%.

Variable	Definition
<i>Panel A: Dependent variables</i>	
$\Delta \log(\text{Emp}_{i,t})$	Log change in number of employees from year $t - 1$ to t : $\log(\text{EMP}_{i,t}) - \log(\text{EMP}_{i,t-1})$. Winsorized at 1%/99%. Dependent variable in Tables 4, 5, 6, 7, 9 and 10.
n_R/n_S (rough separation share)	Share of employee separations at firm i in year t classified as <i>rough</i> : n_R/n_S , where $n_S = n_R + n_{Sm}$ is the number of separations recorded in Revelio Labs <i>Positions</i> data. A separation is rough if the employee’s record shows no subsequent position starting within 90 days of the end date, and smooth (n_{Sm}) otherwise; the 90-day gap proxies for involuntary exit, but a gap of that length need not be involuntary, and Revelio coverage is tilted toward positions visible in online professional records. Column (2) of Table 8 uses the equivalent rough-minus-smooth transform $(n_R - n_{Sm})/n_S = 2(n_R/n_S) - 1$, whose coefficients are therefore exactly twice Column (1)’s with identical t -statistics. Estimated on the Table 4 sample further restricted to firm-years with Revelio separation records. Dependent variables in Table 8.
<i>Panel B: Key independent variables</i>	
$\Delta \text{ROA}_{i,t}$	Change in return on assets: $\text{ROA}_t - \text{ROA}_{t-1}$, where $\text{ROA} = \text{NI}/\text{AT}$. Winsorized at 1%/99%.
$\text{Indiv}_{i,z}$	Firm-year CEO individualism score. Only the CEO’s answers in the Q&A section of the call enter, and within those only the sentences that survive the sentence-level filters; each is scored by its cosine similarity to validated individualism-versus-collectivism anchor sentences, net of a first-person pronoun direction, so that the score is not a restatement of pronoun frequency (contrast $\text{CH}_{i,z}$ below). Q&A answers are delivered without a prepared script, and so less filtered than the remarks that open the same call, which are excluded and scored separately as $\text{Indiv}^{\text{Prep}}$ (Table 11). Answers to analyst questions that request modeling or numerical detail, and to purely clarifying or procedural questions, are also excluded, on the reasoning that in such answers the content is set by the question—a guidance figure, a definition, a handoff to the CFO—rather than by how the CEO attributes authority. Sentence scores are aggregated to the transcript record, then to the mean over each identified speaker’s records within a firm and calendar year—the speaker-firm-year unit of Table 1, Steps 3 and 4, on which the within-CEO stability test of Table 2, Panel A, is computed—as described in the Data and Method section. Where a firm-year carries more than one identified speaker, its value is the score of the speaker appearing on the most records that year, which need not be the CEO in office at the fiscal year end. Standardized within the estimation sample (mean 0, std 1), and not winsorized in any specification except in the section-specific forms of Table 11, whose notes winsorize all continuous variables.

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Table 13 continued

Variable	Definition
$\text{Indiv}_{i,z}^{\text{QA}}, \text{Indiv}_{i,z}^{\text{Prep}}$	The individualism score computed separately on the Q&A section and on the prepared-remarks section of the same calls (written $\text{Indiv}_{i,t,z}^{\text{QA}}$ and $\text{Indiv}_{i,t,z}^{\text{Prep}}$ in Table 11). Indiv^{QA} is the Q&A score of $\text{Indiv}_{i,z}$ above, re-standardized within each estimation sample; $\text{Indiv}^{\text{Prep}}$ is its scripted, investor-relations counterpart, scored on all prepared-remarks sentences, since the screen on analyst question type applies to answers only and has no analogue in prepared remarks. Both serve as dependent variables in Table 11, and $\text{Indiv}^{\text{Prep}}$ enters as a channel control in Table 9.
$\text{Hist}_{i,z}^{\text{QA}}, \text{Hist}_{i,z}^{\text{Prep}}$	Predetermined firm-history terms (written $\text{hist Q\&A}_{i,c}$ and $\text{hist Prep}_{i,c}$ in Table 11): call-weighted averages of the corresponding score over <i>strictly prior</i> CEOs at the same firm i , taken as of the current CEO c , then standardized. Firms with no observed prior CEO are dropped. Used in Tables 11 and 9.
$\Delta\text{ROA}_{i,t} \times \text{Indiv}_{i,z}$	Interaction of the performance shock with CEO individualism. Coefficient β_2 tests whether labor outcomes are more responsive to ROA shocks for individualistic CEOs.
$\Delta\text{ROA}_{i,t}^-, \Delta\text{ROA}_{i,t}^+$	Non-positive and non-negative parts of the performance shock, $\Delta\text{ROA}^- = \min(\Delta\text{ROA}, 0)$ and $\Delta\text{ROA}^+ = \max(\Delta\text{ROA}, 0)$, used in the asymmetric-shock specification of Table 5, Column (2); in the downside columns of Tables 6 and 7, which enter ΔROA^- only; and in both columns of Table 8, which enter both parts.
$\text{CH}_{i,z}$	Standardized Chatterjee and Hambrick (2007) first-person pronoun ratio $I/(I+We)$. It serves as the discriminant-validity benchmark in Table 2, Panel B, where it is measured at the CEO-quarter level, and as the horse-race benchmark in Table 4, Column (5), where it is aggregated to the firm-year level.

Panel C: Firm-level controls (lagged)

$\log(\text{AT})_{t-1}$	Log of total assets at $t-1$. Firm size proxy.
ROA_{t-1}	Lagged return on assets, $\text{NI}_{t-1}/\text{AT}_{t-1}$. Captures profitability persistence and mean reversion.
Leverage_{t-1}	$(\text{DLT} + \text{DLC})/\text{AT}$ at $t-1$.
Cash_{t-1}	Cash and short-term investments, CHE/AT at $t-1$.
SalesGrowth_{t-1}	Lagged sales growth, $(\text{SALE}_{t-1} - \text{SALE}_{t-2})/\text{SALE}_{t-2}$.
$\text{LaborIntensity}_{t-1}$	Labor-to-capital ratio, $\log(\text{EMP}_{t-1}/\text{PPE}_{t-1})$.
$\text{Tobin } Q_{t-1}$	$(\text{PRCC}_F \times \text{CSHO} + \text{AT} - \text{CEQ})/\text{AT}$ at $t-1$.
ELS_{t-1}	Lagged employment labor share at $t-1$, $\text{ELS} = \text{LABEX}/\text{VA}$, where $\text{VA} = \text{OIBDP} + \Delta\text{INVFG} + \text{LABEX}$ (with ΔINVFG set to zero if missing) and LABEX is the FF48 industry-year average wage multiplied by average employment. ELS truncated to (0, 1), winsorized at 1%/99%. Used as a control in the $\Delta \log(\text{Emp})$ regressions.

Panel D: CEO controls (contemporaneous)

CEO Tenure	Number of fiscal years from the executive's first appearance as CEO at firm i in ExecuComp through year t , inclusive, computed for each ($gvkey$, $execid$) pair flagged $\text{ceoann} = \text{CEO}$ as $t - \min(t) + 1$ over that pair's CEO history at the firm. I use this appearance-based measure rather than the BECAMECEO date, which is missing for most firm-years in the CRSP-Compustat-ExecuComp master panel. Where more than one executive is flagged CEO in the same firm-year, the larger value is used. The appearance-based measure is in turn left-truncated at the start of ExecuComp coverage, and so understates the tenure of CEOs appointed before coverage begins.
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Table 13 continued

Variable	Definition
CEO Equity Pay	Share of equity-based pay in total CEO compensation: $(\text{Stock Awards FV} + \text{Option Awards FV}) / (\text{Salary} + \text{Bonus} + \text{Stock Awards FV} + \text{Option Awards FV} + \text{Other})$.
CEO Ownership	CEO ownership of common shares as a percentage of total shares outstanding (<i>shown_tot_pct</i>).

Appendix D Anchor Sentences

Table 14 lists the 100 anchor sentences from which the individualism axis is built.

Table 14: Anchor Sentences

Notes. The 100 anchor sentences loaded by the scoring pipeline. They cover five situations that recur on earnings calls, ten sentences per pole in each. Each pole vector is the equal-weighted mean of its five context sub-centroids; the axis is the difference between the two pole vectors, residualized against a first-person pronoun direction (Section 3). Every sentence names a decision, judgment, responsibility, or expectation and assigns it either to a single individual or to a collective, with no evaluative language about the quality or the outcome of the act, and all are screened against the contamination blacklist of Section 3. This table is generated directly from the anchor file the scorer reads, so the sentences printed here are the sentences used.

<i>Individual pole</i>	<i>Collective pole</i>
<i>Panel A: Performance deviation</i>	
1 My initial assessment of the demand trajectory turned out to be different from what occurred.	The conditions in this area developed differently than the organization had anticipated.
2 The call I made on inventory positioning did not match the conditions that developed.	The assumptions embedded in our plan for this period did not hold.
3 My judgment on the pace of change in this area was not aligned with what actually happened.	The trajectory in this part of the business was different from what our process projected.
4 The approach I selected for this period was based on assumptions that did not hold.	The environment in this area shifted in ways that were not reflected in our internal view.
5 My read on the timing here led me to a position that did not match the environment.	The factors that drove the deviation from plan were not ones that our process had weighted heavily.
6 The decision I made on resource allocation was based on a view that proved to be off.	The outcome in this area differed from what the organization had projected.
7 My assessment of the competitive dynamic led me to an expectation that was not met.	Our collective view on the direction here was not aligned with what actually developed.
8 The priorities I set for this period were based on a view that turned out to differ.	The inputs our process relied on for this area pointed in a different direction than what occurred.
9 My conclusion about the trajectory in this area was not consistent with what unfolded.	The shared assumptions within the organization for this period were not borne out.
10 The direction I set was based on assumptions that I subsequently had to revise.	The organization's position on the trajectory here was different from what the period produced.
<i>Panel B: Major transactions</i>	
1 I identified this as a situation worth pursuing and directed the organization to act.	The decision to pursue this transaction came out of our internal evaluation process.
2 The decision to initiate this transaction was one I made based on my assessment.	Our organization worked through the relevant considerations before committing to this.
3 I set the parameters within which this transaction would be pursued.	The determination to move forward here reflected alignment across the leadership group.
4 My view on the strategic logic here drove the decision to move forward.	We evaluated the factors involved as a group before deciding on a direction.

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Table 14 continued

<i>Individual pole</i>	<i>Collective pole</i>
5 I made the call on whether and when to proceed with this transaction.	The process through which we reached this decision involved input from multiple parts of the organization.
6 The decision to engage in this transaction reflected my judgment on the situation.	Our collective assessment of the situation led us to the conclusion to proceed.
7 I determined that this was the direction the organization should pursue.	The decision on this was made through our established process for evaluating these situations.
8 My assessment of the relevant factors led me to initiate this process.	We organized our analysis across functions before settling on a direction here.
9 I set the terms within which this transaction was negotiated.	The conclusion to move forward reflected a shared view across the relevant teams.
10 The call to commit to this path was mine to make and I made it.	Our organization worked through this question collectively before reaching a position.
<i>Panel C: Strategic redirection</i>	
1 I set the direction the organization would pursue in this area.	The direction being pursued here came out of our internal process for these decisions.
2 The strategic position we are taking here reflects the judgment I applied to the situation.	Our organization worked through the question of how to reorient before setting a direction.
3 My assessment of where this needs to go drove the decision on direction.	The strategic position we are taking reflects input from across the leadership group.
4 I determined that the prior approach needed to be changed and set a new one.	We worked through the relevant considerations as a group before settling on a direction.
5 The framing I applied to this situation led to the direction being pursued.	The approach being pursued reflects a view that developed across the organization.
6 I decided how the organization would reorient itself on this question.	Our collective assessment of what the situation required shaped the direction we set.
7 My view on what the situation required shaped the strategic choice that was made.	We aligned internally on the direction before committing the organization to it.
8 The priorities being pursued here reflect the direction I established.	The strategic choice here reflects a conclusion the organization reached through its process.
9 I concluded that a change in approach was needed and directed the organization accordingly.	We worked through the tradeoffs as a group before establishing a direction.
10 The strategic position we hold reflects a judgment I made about the situation.	The position we are pursuing reflects a shared view developed internally.
<i>Panel D: External challenges</i>	
1 My position on this question is different from the one implied in that framing.	Our position on this question is different from the one implied in the framing.
2 I would frame this issue differently than how it has been presented.	The way the organization thinks about this leads to a different set of conclusions.
3 My read on the relevant factors here leads me to a different conclusion.	Our collective read on the relevant factors points in a different direction.
4 My view on this differs from the assumption embedded in the question.	The view we hold internally on this differs from the assumption in the question.
5 I weight the considerations here differently than the framing of the question implies.	Our shared assessment of the situation leads to a different conclusion.
6 The way I think about this leads me to a different set of conclusions.	How we as an organization approach this question leads to a different set of priorities.

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Table 14 continued

<i>Individual pole</i>	<i>Collective pole</i>
7 My assessment of the situation points in a different direction.	The position the organization holds on this is different from the one suggested.
8 I hold a different position on this than the one being suggested.	Our collective interpretation of the factors leads us to a different place.
9 My interpretation of the relevant factors leads me to a different place.	The conclusion the organization reaches on this is different from the one implied.
10 The conclusion I reach on this question is different from the one implied.	Our internal view on this differs from the framing presented in the question.
<i>Panel E: Forward outlook</i>	
1 My assessment of the trajectory here shapes the position I have communicated.	The outlook we have communicated reflects input gathered from across the organization.
2 The outlook I have set reflects my judgment of the relevant factors.	The position we hold on the trajectory here came out of our internal planning process.
3 My read on the direction of this area is what underpins the position I have taken.	Our collective assessment of the relevant factors underpins the direction we have indicated.
4 I set the parameters for this outlook based on my assessment of the situation.	The targets we have set reflect a view developed across the leadership group.
5 My view on how this will develop is what drives the position I have stated.	Our shared read on the situation drives the outlook we have communicated.
6 The targets I have established reflect my judgment of what the situation warrants.	The parameters of the outlook reflect a collective assessment of the relevant factors.
7 My assessment of the relevant factors is what underpins the direction I have indicated.	We developed the position on this through our internal process.
8 I determined the parameters of this outlook based on my read of the situation.	The direction we have indicated reflects a view that was developed collectively.
9 The position I have taken on this reflects my own assessment of the factors.	Our organization's assessment of the trajectory underpins the position we have communicated.
10 My judgment on the relevant considerations is what drives the outlook I have communicated.	The outlook reflects a shared view developed across the relevant teams.